



Investigating soil moisture–climate interactions in a changing climate: A review

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ABSTRACT

Soil moisture is a key variable of the climate system. It constrains plant transpiration and photosynthesis in several regions of the world, with consequent impacts on the water, energy and biogeochemical cycles. Moreover it is a storage component for precipitation and radiation anomalies, inducing persistence in the climate system. Finally, it is involved in a number of feedbacks at the local, regional and global scales, and plays a major role in climate-change projections. In this review, we provide a synthesis of past research on the role of soil moisture for the climate system, based both on modelling and observational studies. We focus on soil moisture–temperature and soil moisture–precipitation feedbacks, and their possible modifications with climate change. We also highlight further impacts of soil moisture on climate, and the state of research regarding the validation of the relevant processes.

There are promises for major advances in this research field in coming years thanks to the development of new validation datasets and multi-model initiatives. However, the availability of ground observations continues to be critical in limiting progress and should therefore strongly be fostered at the international level. Exchanges across disciplines will also be essential for bridging current knowledge gaps in this field. This is of key importance given the manifold impacts of soil moisture on climate, and their relevance for climate-change projections. A better understanding and quantification of the relevant processes would significantly help to reduce uncertainties in future-climate scenarios, in particular with regard to changes in climate variability and extreme events, as well as ecosystem and agricultural impacts.

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1. Introduction

The water stored on land is a key variable controlling numerous processes and feedback loops within the climate system. It is of major relevance for the water and energy cycles, but also impacts the exchanges of trace gases on land, including carbon dioxide. Fig. 1 provides an overview of the main components and exchanges within the climate system. The extreme complexity of land processes and feedbacks is apparent, in particular if one considers the strong heterogeneity of the land cover and its temporal variability.

Soil moisture generally refers to the amount of water stored in the unsaturated soil zone, although its exact definition can vary depending on the context, i.e. whether it is defined in relative, absolute or indirect terms, and depending on the reference storage (see Section 2). Soil moisture is a source of water for the atmosphere through processes leading to evapotranspiration from land, which include mainly plant transpiration and bare soil evaporation. Evapotranspiration is a major component of the continental water cycle, since it returns as much as 60% of the whole land precipitation back to the atmosphere (e.g. Oki and Kanae, 2006). In addition, evapotranspiration is also an important energy flux: The energy required for evaporating a given mass (e.g. 1 g) of liquid water corresponds to approximately 600 times the energy required to increase its temperature by 1 K, and to 2400 times the energy required to increase the temperature of a corresponding mass of air by 1 K. Consequently, land evapotranspiration also uses up more than half of the total solar energy absorbed by land surfaces (Trenberth et al., 2009). Through its impact on the partitioning of the incoming energy in the latent and sensible heat fluxes, soil moisture has several additional impacts on climate processes, in particular on air temperature, boundary-layer stability and in some instances on precipitation.

While soil moisture–climate interactions have received increasing interest in recent years, the complexity of the involved feedbacks and the scarcity of observations for characterizing the relevant processes have also become increasingly apparent. Nonetheless, some promising perspectives are opening in this field, both linked to the modelling, validation and monitoring of the relevant processes, and to the development of related key applications such as short-term and seasonal forecasting, drought assessment and ecosystem management.

In this review, we present a synthesis of past research on the role of soil moisture for the climate system, and discuss current perspectives for research in this field. We focus here on physical aspects linked to the water and energy cycles in the context of climate change. The article is structured as follows: We first provide some basic definitions for soil moisture (Section 2). In Section 3, we discuss the role of soil moisture for the land energy and water balances through its impact on evapotranspiration. In Section 4, we provide an overview on soil moisture–evapotranspiration coupling and its representation in current models. Section 5 discusses the main resulting impacts on temperature and precipitation, and Section 6 addresses further relevant aspects. Section 7 presents the possible modifications of land–climate interactions with climate change. In Section 8, we present methods available to measure or estimate soil moisture and evapotranspiration, and describe the current state of research regarding validation datasets for these two quantities. Finally, we discuss current issues in this field and perspectives for future research in Section 9, and provide the main conclusions in Section 10. An overview of the terminology followed in this review regarding the definition of the terms “coupling”, “feedback” and “interactions” is provided in the Appendix A. Relevant coupling diagnostics, which will be referred to in several sections, are introduced in the Appendix B.

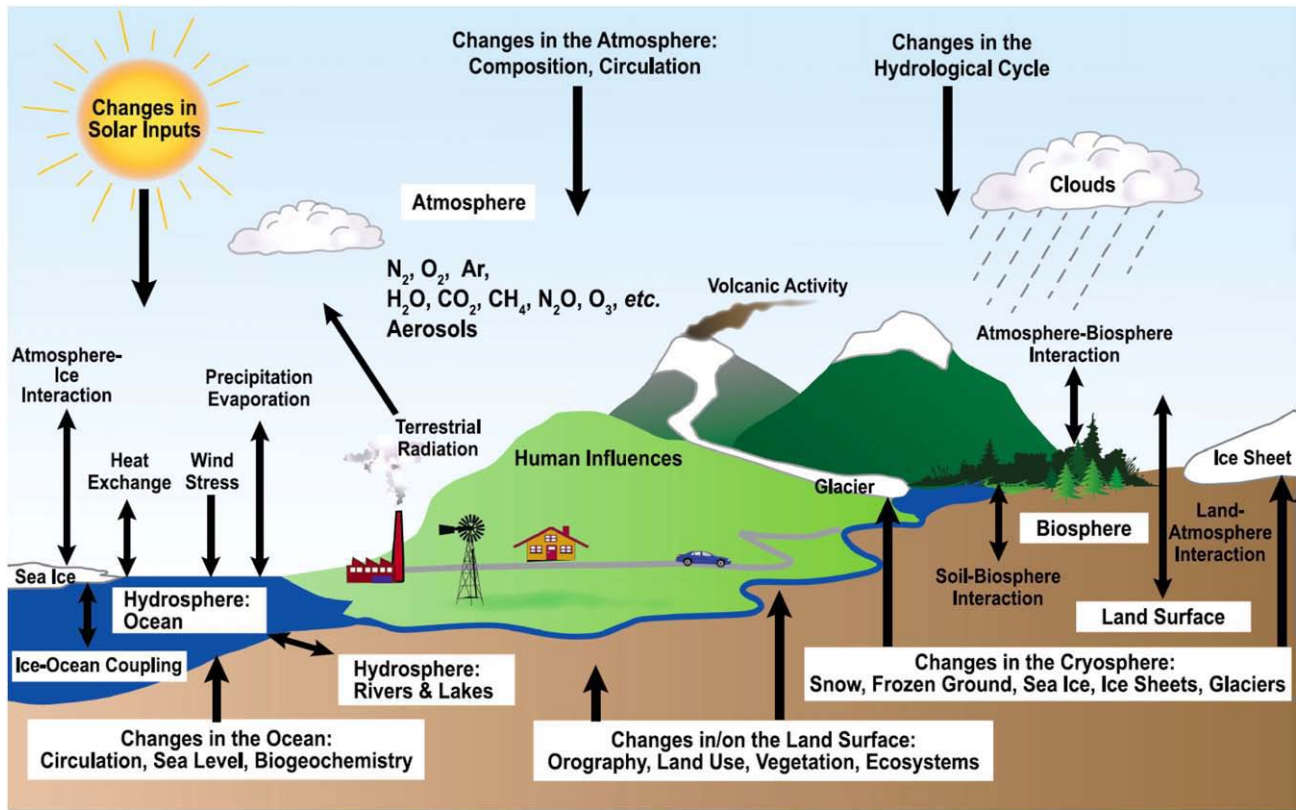


Fig. 1. Global climate system [figure from IPCC, 2007].

2. Soil moisture: Some definitions

Soil moisture is usually defined as the water contained in the unsaturated soil zone (e.g. Hillel, 1998), also termed vadose zone (Fig. 2). In practice, often only a fraction of soil moisture is relevant or measurable. Thus, soil moisture needs to be considered with regard to a given soil volume.

Volumetric soil moisture θ [$m^3 H_2O/m^3_{soil}$] in the soil volume V (e.g. volumes A or B in Fig. 2) is defined as follows:

$$\theta = \left(\frac{\text{volume of water in } V}{V} \right) \quad (1)$$

Note that in principle, Eq. (1) is applicable on multiple scales, from a few cubic centimeters to several cubic kilometers, depending on the measurement method or research application. In land surface and hydrological models, for which the soil is divided in discrete soil layers, θ is often expressed as [mm_{H_2O}/mm_{soil}] (Fig. 3).

One conceptual issue with the definition of soil moisture is the characterization of the soil volume V in Eq. (1). Indeed, soil moisture content is not homogeneously distributed vertically and horizontally, and thus differs for different soil volumes. This is of high relevance for the comparison of measurements/estimation methods (Section 8), which may for instance only provide estimates of soil moisture in the top few centimeters of the soil (e.g. microwave remote sensing), in a small volume at a given depth (e.g. TDR measurements), or together with (and undistinguished from) other parts of terrestrial water storage (snow, groundwater, surface water), as is the case for GRACE or BSWB estimates (Section 8). Moreover, for certain soil moisture definitions (e.g. root-zone soil moisture, total soil moisture), V may be a function of space and time (i.e. as a function of the plants' rooting depth, e.g. Jackson et al., 1996; Schenk and Jackson, 2002, or the water table depth).

Another important quantity is the actual maximum soil moisture content of the given soil volume, i.e. the porosity. This maximum soil moisture content is defined as the saturation soil moisture content,

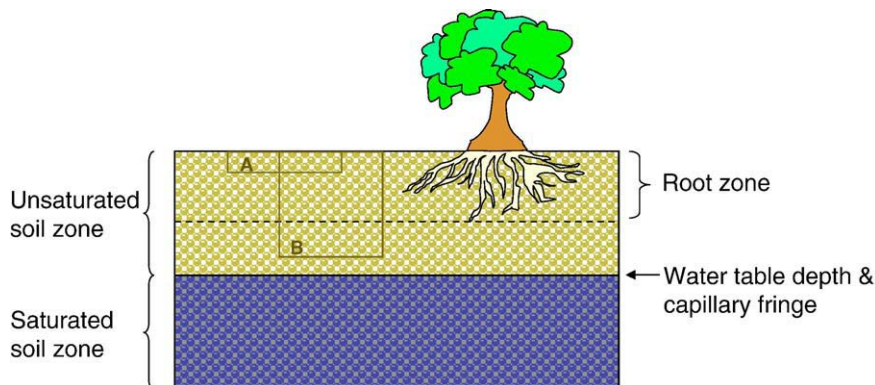


Fig. 2. The saturated and unsaturated soil zones. A and B denote two distinct soil moisture volumes.

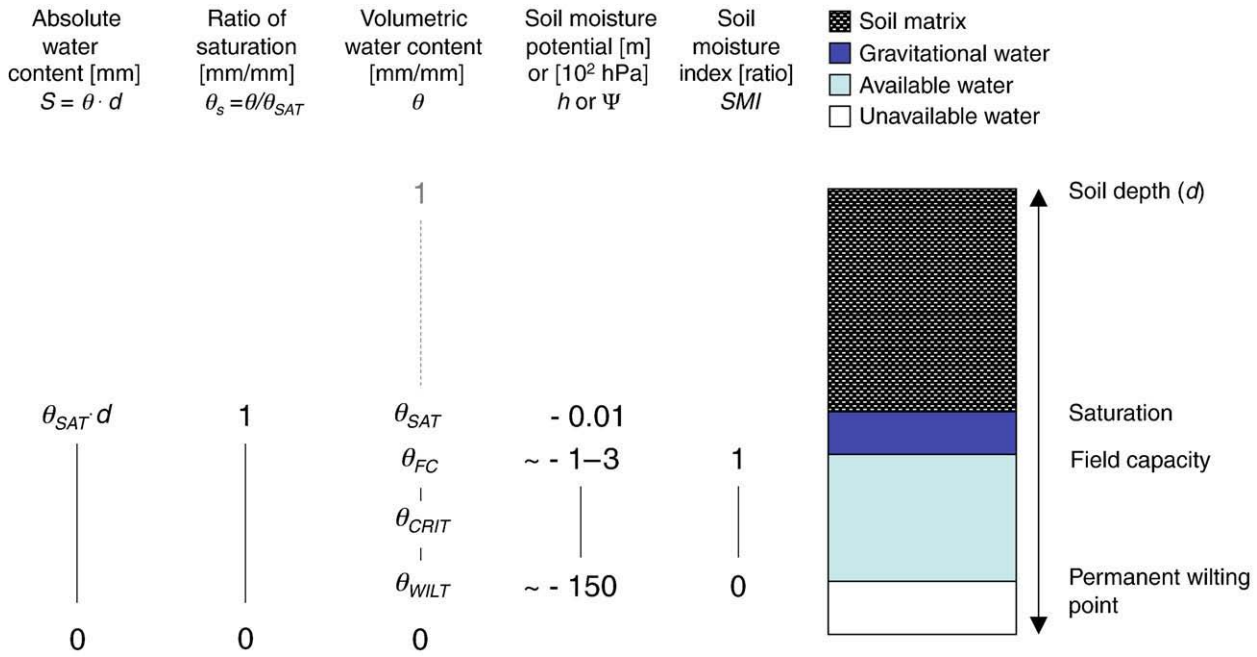


Fig. 3. Characteristic soil moisture levels and units. The schematic on the right represents a given soil sample of depth d (and uniform cross section), with typical soil moisture levels. At saturation θ_{SAT} , all pores are filled with water, while at the plant wilting point θ_{WILT} , only the water strongly bound to the soil matrix (and thus inaccessible to plants) is remaining. Note that the soil moisture potential values (expressed as pressure head h or soil moisture tension/soil's matrix suction ψ) are approximate. See text for more details.

or θ_{SAT} (also termed water-holding capacity). Using θ_{SAT} , a further common definition of soil moisture is the saturation ratio:

$$\theta_s = \frac{\theta}{\theta_{SAT}} \quad (2)$$

with θ_s varying between 0 (no soil moisture) and 1 (full saturation), while θ varies between 0 and θ_{SAT} (≤ 1 ; see Fig. 3).¹

Note that in reality, the maximum volume of water available to plants is the *field capacity* (θ_{FC}) minus the *permanent wilting point* (θ_{WILT})² (Fig. 3). Above the field capacity, water cannot be held against gravitational drainage, and below the wilting point, the water is held too strongly by the soil matrix and is not accessible to plants (for more details, see e.g. Hillel, 1998). The field capacity and the wilting point depend on soil properties such as soil texture (and on vegetation type for the wilting point, Sperry et al., 2002; Hupet et al., 2005, see also below), and thus vary geographically. Using these two quantities, an additional soil moisture definition can be proposed, usually termed “soil moisture index” (SMI, e.g. Betts, 2004; see also Fig. 3):

$$SMI = \frac{\theta - \theta_{WILT}}{\theta_{FC} - \theta_{WILT}} \quad (3)$$

The SMI is thus a measure of soil moisture content as ratio of the total storage available to plants (varying between 0 and 1). Note that in certain circumstances actual soil moisture can be above the field capacity (e.g. after heavy precipitation events or in the saturated areas alongside rivers). To avoid this issue, the SMI can be set to 1 for any values above θ_{FC} .

While Eqs. (1)–(3) express soil moisture in relative terms, i.e. as ratio of a given soil volume or water storage [m^3/m^3] or [mm/mm], soil moisture can also be defined in absolute terms (water depth [mm] or mass [kg]). This is particularly relevant for the computation of the terrestrial water balance (Section 3, Eq. (5)). In this case, absolute soil

moisture content is often expressed as S (soil moisture [mm], see Fig. 3), with:

$$S = \theta \cdot d \quad (4)$$

where d is the considered soil depth (with constant cross section).

Finally, it is also useful to express soil moisture as soil moisture potential (soil tension or soil's matrix suction, see Fig. 3). This quantity is essential for characterizing the binding of soil moisture to the soil matrix. The saturation water content θ_{SAT} , the field capacity θ_{FC} , and the permanent wilting point θ_{WILT} , are typically defined as corresponding to soil moisture potentials of -1 hPa, ca. -100 – 300 hPa, and ca. $-15,000$ hPa (1.5 MPa), respectively. Corresponding suction heads (negative pressure heads) equal 1 cm, 1–3 m and 150 m, respectively (Fig. 3). Note that these values are approximate (see e.g. Shaw, 1994; Hillel, 1998; Dingman, 2002), and, for the plant wilting point, also depend on the vegetation type (Sperry et al., 2002; Hupet et al., 2005). For instance, temperate oaks can extract water at potentials as low as -2 MPa (Bréda et al., 1995), while Mediterranean tree or shrub species can extract water at potentials as low as -4 or -5 MPa (Damesin and Rambal, 1995; Rambal et al., 2003). However, the potentials for θ_{SAT} , θ_{FC} , and θ_{WILT} are about equivalent for different soil types, while the corresponding volumetric soil moisture contents strongly depend on the soil texture (e.g. Hillel, 1998).

These distinct soil moisture definitions are of importance in the context of ground measurements, as often only one of them can be measured or indirectly estimated. This is particularly critical because measurements of soil moisture are mostly indirect (e.g. dielectric constant, variations of gravity field, or soil suction), and are generally related only to one specific definition of soil moisture (see also Section 8).

When concerned about feedbacks of soil moisture content to climate, the main quantity of interest is the soil moisture storage

¹ Obviously, these constitute theoretical limits, since values of $\theta = \theta_s = 0$ can only be reached in the laboratory and are not found in the field (see further).

² Also called *plant wilting point*.

³ This common definition can, however, correspond to markedly distinct absolute soil moisture contents in different regions (e.g. Jackson et al., 1996; Schenk and Jackson, 2002), as well as in different land surface models for the same region (e.g. Dirmeyer et al., 1999; Seneviratne et al., 2006b; Koster et al., 2009b; Wang et al., 2009).

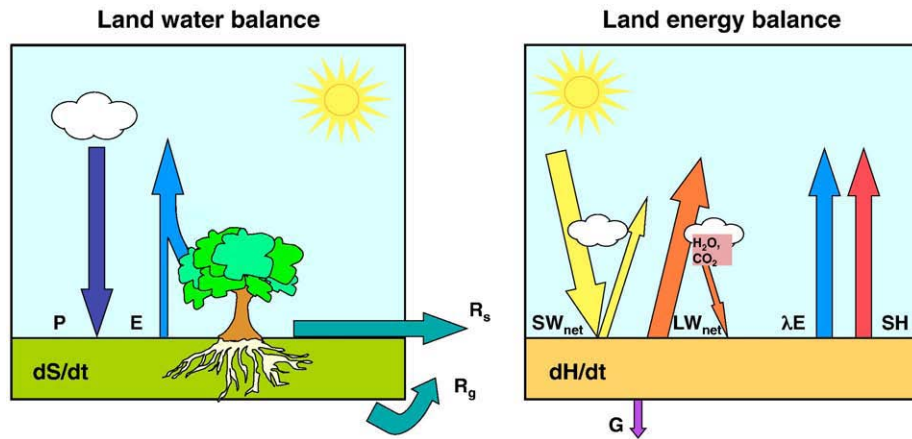


Fig. 4. Schematic of the land water balance (left) and land energy balance (right) for a given surface soil layer. dS/dt refers to the change in water content within the layer (soil moisture, surface water, snow; depending on the depth of the layer, this may include ground water changes), while dH/dt refers to the change of energy within the same layer. SW_{net} refers to the net shortwave radiation ($SW_{in} - SW_{out}$) and LW_{net} refers to the net longwave radiation ($LW_{in} - LW_{out}$). Note that H_2O and CO_2 refer to atmospheric water vapour and atmospheric CO_2 and their role as greenhouse gases. For simplicity other greenhouse gases are not indicated on the figure. For explanations of the other abbreviations, see Section 3.

affecting plants' transpiration, since transpiration is overall the largest contributor to total land evapotranspiration (e.g. Dirmeyer et al., 2006a; Lawrence et al., 2007). Generally, it is equivalent to the soil moisture content and corresponding soil moisture potential within the root zone.³ In the present review, this is the definition we will assume for soil moisture whenever no additional specification is provided. Note nonetheless that surface soil moisture can be more relevant in some regions, e.g. due to its impact on surface runoff or on bare soil evaporation (see Section 4.2).

3. The role of soil moisture in the land energy and water balances

Before addressing the impact of soil moisture on specific climate processes (Sections 4–6), we provide here an overview on its role in the context of the land energy and water balances (Fig. 4). Soil moisture plays an important part in both budgets through its impact on evapotranspiration E (or latent heat flux λE , where λ is the latent heat of vaporization).

The land water balance for a surface soil layer including vegetation (Fig. 4, left) without considering lateral exchange between adjacent soil volumes can be expressed as:

$$\frac{dS}{dt} = P - E - R_s - R_g \quad (5)$$

where dS/dt is the change of water content within the given layer, P is the precipitation, E is the evapotranspiration, R_s is the surface runoff, and R_g is the drainage. Note that the term dS/dt includes soil moisture (absolute content, see Section 2 and Eq. (4)), but can also encompass additional forms of water storage, such as surface water, snow, ice cover, and – depending on the depth of the considered soil layer – groundwater (in the latter case, R_g corresponds to the baseflow). It also includes the interception storage. The evapotranspiration term E includes bare soil evaporation, plants' transpiration, evaporation from the interception storage, snow sublimation, and evaporation from surface water.

The land energy balance for the same surface soil layer (Fig. 4, right) can be expressed as:

$$\frac{dH}{dt} = R_n - \lambda E - SH - G \quad (6)$$

³ If E includes snow sublimation, then λ refers to the latent heat of sublimation for the sublimation term.

where dH/dt is the change of energy within the considered surface soil layer (including vegetation) assumed to include all water storage components considered in Eq. (5) (i.e. temperature change, phase changes associated with soil freezing/melting or snow melt), R_n is the net radiation, λE is the latent heat flux,⁴ SH is the sensible heat flux, and G is the ground heat flux to deeper layers. Note that for an infinitesimally small layer, dH/dt tends to zero and G is the ground heat flux at the surface.

The net radiation R_n is defined as follows:

$$R_n = SW_{in} - SW_{out} + LW_{in} - LW_{out} \quad (7)$$

where SW_{in} is the incoming shortwave radiation, SW_{out} is the outgoing shortwave radiation (equal to $SW_{in} \cdot \alpha$, where α is the albedo), LW_{in} is the incoming longwave radiation, and LW_{out} is the outgoing longwave radiation.

From Eqs. (5) and (6), one can easily see that the land energy and water balances are coupled through the evapotranspiration term (E , λE). Thus soil moisture plays a key role both for the water and energy cycles (through its impact on the energy partitioning at the surface). In addition, it is also linked to several biogeochemical cycles (e.g. carbon and nitrogen cycles) through the coupling between plants' transpiration and photosynthesis.

These effects are, however, only important in regions where soil moisture is the main controlling factor for evapotranspiration. We will discuss the relevance of this aspect in the following section.

4. Soil moisture–climate interactions and feedbacks: Soil moisture–evapotranspiration coupling

Most of the inferred impacts of soil moisture for the climate system are induced by its role for evapotranspiration in soil moisture-limited regimes. The soil moisture content is directly linked to the soil water potential or soil suction, i.e. how tightly the water is bound to the soil matrix (Section 2). With decreasing soil moisture content, the soil suction increases, the remaining soil moisture becomes less accessible for uptake by plant roots, and evapotranspiration might thus become reduced. A number of processes are relevant for the relationship between soil moisture and evapotranspiration.

In this subsection we provide an overview of the current state of knowledge regarding the dependency of evapotranspiration on soil moisture. First we discuss the classical framework followed for the definition of evapotranspiration regimes (Section 4.1). Then we provide an overview on how this relationship is represented in current land surface models (Section 4.2). Inferences regarding the location of

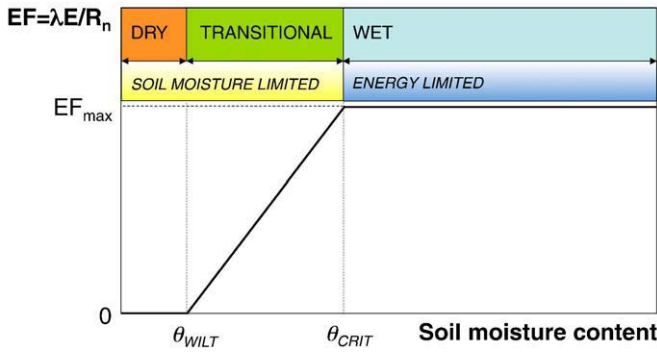


Fig. 5. Definition of soil moisture regimes and corresponding evapotranspiration regimes according to framework described in Section 4.1. EF denotes the evaporative fraction, and $EF_{\max}$ its maximal value.

regions of strong soil moisture–evapotranspiration coupling based on modelling studies are presented in Section 4.3, and results from observational studies are discussed in Section 4.4. While we focus here on the direct sensitivity of evapotranspiration to soil moisture content, one should keep in mind that further processes contributing to the land water balance (e.g. infiltration, percolation, interception of water on plants, anthropogenic or natural land cover changes) will ultimately also impact this relationship, albeit indirectly.

4.1. Classical conceptual framework

Classical hydrology provides a useful conceptual framework (Fig. 5) for the definition of evapotranspiration regimes as a function of soil moisture, based on the original publications of Budyko (e.g. Budyko, 1956, 1974), and recently highlighted in e.g. Koster et al. (2004a, 2009a), Seneviratne et al. (2006a), and Teuling et al. (2009). Two main evapotranspiration regimes, characterized with the evaporative fraction ($EF = \lambda E / R_n$) can be defined: a soil moisture-limited and an energy-limited evapotranspiration regime. In the energy-limited evapotranspiration regime, corresponding to values of soil moisture lying above a given critical soil moisture value θ_{CRIT} , evaporative fraction is independent of the soil moisture content. Below θ_{CRIT} , soil moisture content provides a first-order constraint on evapotranspiration (soil moisture-limited evapotranspiration regime). The other important threshold is the wilting point θ_{WILT} under which no evapotranspiration takes place anymore (Section 2). Correspondingly, three climate/soil moisture regimes can be defined, related to the impact of soil moisture on evapotranspiration variability (Koster et al., 2004a; Seneviratne et al., 2006a): Wet ($\theta > \theta_{\text{CRIT}}$) and dry ($\theta < \theta_{\text{WILT}}$) climate regimes, where soil moisture does not impact evapotranspiration variability, and a transitional climate regime ($\theta_{\text{WILT}} \leq \theta \leq \theta_{\text{CRIT}}$), where soil moisture strongly constrains evapotranspiration variability and thus resulting feedbacks to the atmosphere.

While several other controls limit evapotranspiration besides soil moisture and the linear dependency is a simple approximation (Section 4.2), recent studies have highlighted that this framework provides a good first-order representation of regimes of land–atmosphere coupling, both in models and observations (e.g. Koster et al., 2004a,b; Seneviratne et al., 2006a; Teuling et al., 2006; see also Sections 4.3, 4.4 and 5). Indeed, even for complex physiological models (Section 4.2), soil moisture provides a first-order control on land–atmosphere exchanges when it is limiting. We will refer back to this framework in the following sections.

4.2. Simulation of evapotranspiration in current climate models

Land surface models (LSMs, also named “Soil–Vegetation–Atmosphere Transfer schemes” (SVATs), “land surface schemes” (LSSs), or “land surface parameterizations”) are modules employed in climate

models to represent land surface processes. They describe the dependency of evapotranspiration on soil moisture in varying degrees of complexity (e.g. Sellers et al., 1997; Arora, 2002; Pitman, 2003). We briefly present the distinctions between commonly used LSMs regarding the representation of evapotranspiration in the following paragraphs.

As discussed in the preceding section, already first conceptual models of land–atmosphere interactions considered the role of soil moisture for evapotranspiration, by distinguishing radiation-limited regimes and soil moisture-limited regimes. The corresponding most simple representation of evapotranspiration in the context of a climate model, is the so-called “bucket model” (Manabe, 1969, based on Budyko, 1956; see also “first-generation schemes” in Sellers et al., 1997). The bucket model defines actual evapotranspiration as the potential evaporation (E_{POT}) multiplied by a limiting factor (“ β factor”, also called evaporation factor or stress factor). The limiting factor β often consists in a simple linear function of soil moisture content between two given thresholds: θ_{CRIT} , above which E equals E_{POT} ; and θ_{WILT} , below which E is zero (e.g. Manabe, 1969; Sellers et al., 1997; Brutsaert, 2005; similar to the approach followed for the definition of the evaporative fraction in Section 4.1., but using E_{POT} instead of R_n/λ as scaling factor). θ_{CRIT} lies between the wilting point θ_{WILT} and the field capacity θ_{FC} , and is typically equal to ca. 50–80% of θ_{FC} (e.g. Shuttleworth, 1993). The resulting expression for evapotranspiration is as follows:

$$E = \beta E_{\text{POT}} = \beta \rho_a \left[\frac{q_{\text{sat}}(T_s) - q_r}{r_a} \right] \quad (8a)$$

with

$$\beta = \frac{\theta - \theta_{\text{WILT}}}{\theta_{\text{CRIT}} - \theta_{\text{WILT}}} \quad \text{for } \theta_{\text{WILT}} \leq \theta \leq \theta_{\text{CRIT}} \quad (8b)$$

$$\begin{aligned} \beta &= 1 \quad \text{for } \theta > \theta_{\text{CRIT}} \\ \beta &= 0 \quad \text{for } \theta < \theta_{\text{WILT}} \end{aligned}$$

where θ is the soil moisture content, ρ_a is the air density, r_a is the aerodynamic resistance between the surface and the lowest layer of the atmosphere, $q_{\text{sat}}(T_s)$ is the saturation water-vapour specific humidity at the temperature of the surface, and q_r is the water-vapour specific humidity at the reference atmospheric level. Because θ_{CRIT} , the critical soil moisture content under which soil moisture becomes limiting, is generally lower than the field capacity (see above), there is a soil moisture regime under which soil moisture does not limit evapotranspiration (wet regime, i.e. $\theta > \theta_{\text{CRIT}}$, see also Section 4.1). Note that a distinction between Eqs. (8a) and (8b) and the original framework of Budyko (Section 4.1) is the use of E_{POT} instead of R_n/λ as scaling factor for the evapotranspiration. However, R_n/λ actually presents some advantages over E_{POT} (Koster et al., 2009a), because the latter is by definition overestimated when $\theta < \theta_{\text{CRIT}}$, being based on the actual surface temperature rather than the “equivalent wet temperature” (see Milly, 1992, for more details).

Eqs. (8a) and (8b), as implemented in the bucket model, represented an important first attempt to capture soil moisture limitation on evapotranspiration in climate models. Nonetheless, several investigations have shown that bucket-type parameterizations are inadequate for a number of reasons, most importantly because they do not include additional aspects limiting plants' transpiration beside soil moisture. This generally results in an overestimation of evapotranspiration compared to other land surface schemes (e.g. Henderson-Sellers et al., 1996; Seneviratne et al., 2002). In addition, even unstressed transpiration tends to be overestimated, as maximum evapotranspiration in reality is lower than E_{POT} (evaporation from an open water surface) due to additional limiting factors associated with plant physiology (e.g. Viterbo, 1996). Furthermore, E_{POT} is itself overestimated in dry regimes

(see discussion of scaling factors E_{TOT} vs R_n/λ above). Other common issues of bucket-type models are that they generally include only one soil storage, do not consider interception storage and geographical variations in soil and vegetation parameters (see below), and have a too simple representation of runoff generation, temperature conduction, and soil freezing (e.g. Mitchell and Warrilow, 1987; Wood et al., 1998; Viterbo et al., 1999; Pitman, 2003).

Biophysical LSMs, also termed “second-generation” models (Sellers et al., 1997), are characterized by an explicit canopy, multi-layer soil, and the accounting of the main physical processes occurring within the vegetation stand and in the soil (e.g. BATS, Dickinson, 1984; SiB, Sellers et al., 1986). This includes evaporation from four distinct sources (potential evaporation from the interception layer, bare soil evaporation, transpiration from the vegetation, snow sublimation), leaf and snow drip, infiltration, and percolation. The vegetation cover can make use of deep root-zone water for transpiration, allowing for long-term climate memory. Indeed, the bucket model has a typical total water-holding capacity of 0.15 m globally (Manabe, 1969; Sellers et al., 1997) compared to values of ca. 0.25 to 1.5 m for 2nd-generation LSMs (Seneviratne et al., 2006b). Surface parameters vary geographically, and are specified using look-up tables for the different land cover classes.

The most important feature of biophysical models is the (biophysical) control on evapotranspiration through the stomatal resistance r_s . The stomatal resistance represents the additional resistance to water transfer imposed by the stomates, apertures in the leaves through which water and CO₂ are exchanged in plants. The computation of r_s often follows the Jarvis (1976) approach (see e.g. Sellers et al., 1997; Dickinson, 1984):

$$r_s = r_{s\min} f(\text{PAR})f(T)f(\delta e)f(\Psi_1) \quad (9)$$

where the factors $f(\text{PAR})$, $f(T)$, $f(\delta e)$, and $f(\Psi_1)$, give the dependence of r_s on solar radiation (photosynthetically active radiation, PAR), temperature, vapour pressure deficit, and leaf water potential (directly linked to soil moisture content), respectively. Under stress-free conditions, all four factors are equal to 1 and r_s is equal to the minimum stomatal resistance $r_{s\min}$ (else r_s is higher).

The stomatal resistance at the leaf level r_s is scaled up to obtain the canopy resistance r_c . A common method is to use the Leaf Area Index (LAI [–], defined as area of leaves [m²] over surface area [m²]), assuming that the leaves' resistances act in parallel (e.g. Dickinson, 1984):

$$r_c = \frac{r_s}{\text{LAI}} \quad (10a)$$

or

$$g_c = g_s \text{LAI} \quad (10b)$$

where g_s is the stomatal conductance ($1/r_s$), and g_c is the canopy conductance ($1/r_c$). Note that this approach can be refined by taking into account the proportion of sunlit and shaded leaves, using a “two-leaf model” (Wang and Leuning, 1998; Diak et al., 2004) instead of the typical “big-leaf” approach.

The resulting canopy transpiration E_{TR} is then defined as follows:

$$E_{\text{TR}} = \rho_a \left[\frac{q_{\text{sat}}(T_s) - q_r}{r_a + r_c} \right] \quad (11)$$

If soil evaporation, evaporation from the interception storage and snow sublimation can be neglected, then total evapotranspiration is equivalent to plants' transpiration in biophysical models, i.e. Eq. (11). In that case, as pointed out by Sellers et al. (1997), one difference between Eq. (11) and Eq. (8a) resides in the fact that the soil moisture limitation is applied externally to potential evaporation in the bucket model (Eq. (8a)), while in biophysical models (Eq. (11)) the chain of resistances impeding evapotranspiration is represented as acting in series. It is, however, possible to reformulate Eq. (11) as an expression

equivalent to Eq. (8a) by setting r_c to be equal to $((1 - \beta)/\beta)r_a$. In this view, the real difference between Eqs. (11) and (8a) rather appears to lie in the following aspects. First, stomatal resistance r_s is a function of several limiting factors, not only soil moisture. Second, r_s can be measured in the field: While one cannot expect that leaf-level measurements (e.g. of $r_{s\min}$) can be used per se as parameters in climate models due to the scale discrepancy (e.g. Blyth et al., 1993) and the range of measurements (e.g. Garratt, 1993), this allows nonetheless a link to an observable, e.g. with respect to relevant controlling factors. In effect, Eq. (9) uses empirical relationships derived from field measurements (Jarvis, 1976).⁵ Finally, the biophysical LSMs include several features impacting evapotranspiration more indirectly (e.g. Koster and Suarez, 1994; Sellers et al., 1997). Among other aspects, one can mention the distinction between bare soil and root-zone evaporation, which allows for two separate storages of soil moisture with distinct memory characteristics and respective impacts on surface fluxes (surface soil moisture being critical in controlling bare soil evaporation and infiltration, e.g. Penman, 1947; Mahfouf and Noilhan, 1991; Kurc and Small, 2004); the geographical details of variations in soil and vegetation parameters, in particular the water-holding capacity and the rooting depth, despite on-going uncertainties regarding their specification (e.g. Nepstad et al., 1994; Jackson et al., 1996; Canadell et al., 1996; Jackson et al., 2000; Kleidon and Heimann, 2000; Feddes et al., 2001; Schenk and Jackson, 2002, 2005; Seneviratne et al., 2006b; Anders and Rockel, 2009); and the inclusion of an interception storage, which allows “fast evaporation”, a very fundamental improvement over bucket models given the importance of this flux in several regions of the world (e.g. Rutter et al., 1975; Gash et al., 1980; Shuttleworth, 1988; Dolman and Gregory, 1992). These can also explain some of the large differences found between results from bucket-type and biophysical LSMs.

More recently, also physiological (in addition to biophysical) aspects have gained more attention (“third-generation models”, Sellers, 1997). In particular, a growing number of LSMs account for plant photosynthesis, including carbon assimilation, nutrients uptake, enzyme kinetics, electron transport, and the light interception by chloroplasts in plant leaves (e.g. Sellers et al., 1996; Bonan, 1996; Foley et al., 1996; Sellers et al., 1997; Cox et al., 1998, 1999; Knorr, 2000; Krinner et al., 2005). Other more empirical approaches are used in some ecological and/or land surface model frameworks (e.g. Arora, 2002). Physiological models also include a soil moisture limitation on photosynthesis, and thus can in principle incorporate observed impacts of soil moisture variations on plant assimilation and respiration (e.g. Ciais et al., 2005; Jones and Cox, 2005; Granier et al., 2007; Reichstein et al., 2007; Schwalm et al., 2009; Phillips et al., 2009). Possible effects of CO₂ concentrations on the water-use efficiency of plants (e.g. Field et al., 1995; Gedney et al., 2006; Leuzinger and Körner, 2007), and thus modifications of the soil moisture–evapotranspiration relationship under elevated CO₂, are also included. However, much debate subsists on the representation of the relevant processes, especially regarding changes in transpiration under elevated CO₂. These uncertainties result in major divergences in climate-change projections between state-of-the-art carbon-cycle climate models (e.g. Cramer et al., 2001; Friedlingstein et al., 2006; Gedney et al., 2006; Piao et al., 2007; Betts et al., 2007; Bonan, 2008). As a consequence, these physiological aspects have not been included in many modelling applications, and are also not part of the standard climate-change projections of the most recent IPCC AR4 report⁶ (Randall et al., 2007; Meehl et al., 2007; Christensen et al.,

⁵ However, another limitation besides scaling issues is the lack of comprehensive observations around the globe to constrain this relationship for different plant types and climate conditions.

⁶ Nonetheless, C4MIP simulations (Friedlingstein et al., 2006) were used for the interpretation of the IPCC AR4 projections (Meehl et al., 2007).

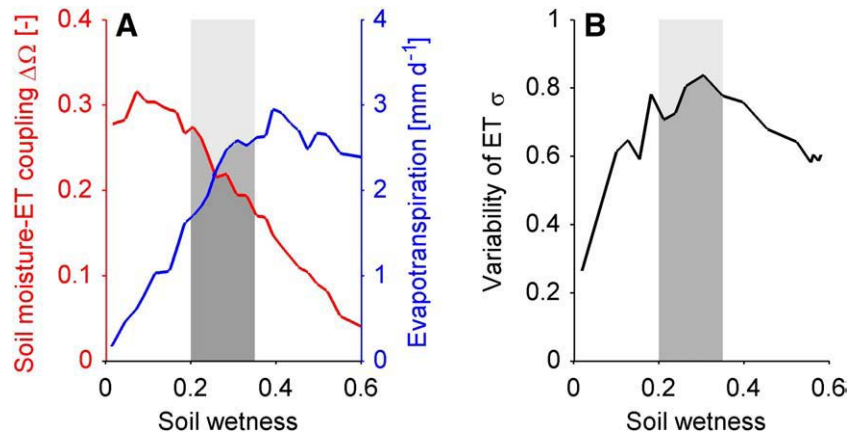


Fig. 6. Dependence on soil wetness of coupling of soil moisture and evapotranspiration estimated with $\Delta\Omega$ diagnostic (left) and standard deviation of evapotranspiration (right) in GLACE simulations (computed from 224 aggregated 6-day totals, corresponding to 16 ensemble members times 14 intervals over analysed JJA period). See Appendix B.1 and Eq. (18) for more details on the $\Delta\Omega$ diagnostic. [Figures adapted from Koster et al., 2004a (left) and Guo et al., 2006a (right)].

2007). It is planned that these processes will be included in some global climate model simulations of the IPCC AR5.

4.3. Regions of strong soil moisture–evapotranspiration coupling: Modelling synthesis

We have seen in the previous subsection that evapotranspiration can be represented with varying degrees of complexity in LSMs. This results in distinct relationships between soil moisture and evapotranspiration in the corresponding models. One may, however, reflect on whether an overall picture can be derived on these interactions independently of individual model details.

The GLACE experiment (Koster et al., 2004a, 2006; Guo et al., 2006a) investigated land–atmosphere coupling characteristics of 12 state-of-art Atmospheric General Circulation Models (AGCMs). Despite notable differences between the models, some common features could be determined. In particular, geographical hot spots of land–atmosphere coupling, common to many of the investigated AGCMs, were identified. These regions were found to be located in transitional zones between dry and wet climates (Koster et al., 2004a).

The explanation for this result is directly related to the dependency of evapotranspiration (mean and variability) on soil moisture in the

models, which is very reminiscent of the classical conceptual framework illustrated in Fig. 5: As displayed in Fig. 6 (based on analyses from Koster et al., 2004a; Guo et al., 2006a), in dry regions, the models' evapotranspiration is strongly controlled by soil moisture (red line in left panel), but its absolute value (blue line) and variations (right panel) are too small to impact climate variability. In wet regions, the models' evapotranspiration is large, but is not controlled by soil moisture, thus soil moisture has little impact on evapotranspiration. Only in transitional regions between dry and wet climates are both conditions met for strong soil moisture–climate coupling: a strong dependency of evapotranspiration on soil moisture and large mean evapotranspiration (and variations thereof, Fig. 6 right). We will discuss implications of these findings for soil moisture–temperature and soil moisture–precipitation interactions in Sections 5.1 and 5.2, respectively.

Interestingly, the derived relationship between soil moisture and evapotranspiration in Fig. 6 (blue line in left panel) appears strikingly similar to that implied by Eqs. (8a) and (8b) and the classical framework of Fig. 5, i.e. a linear relationship between two given thresholds. One should note that this applies to the mean behaviour of the models and not to each model individually (e.g. Dirmeyer et al., 2006b). Nonetheless, this shows that some of the earliest assumptions on soil moisture–evapotranspiration coupling (e.g. Budyko, 1956;

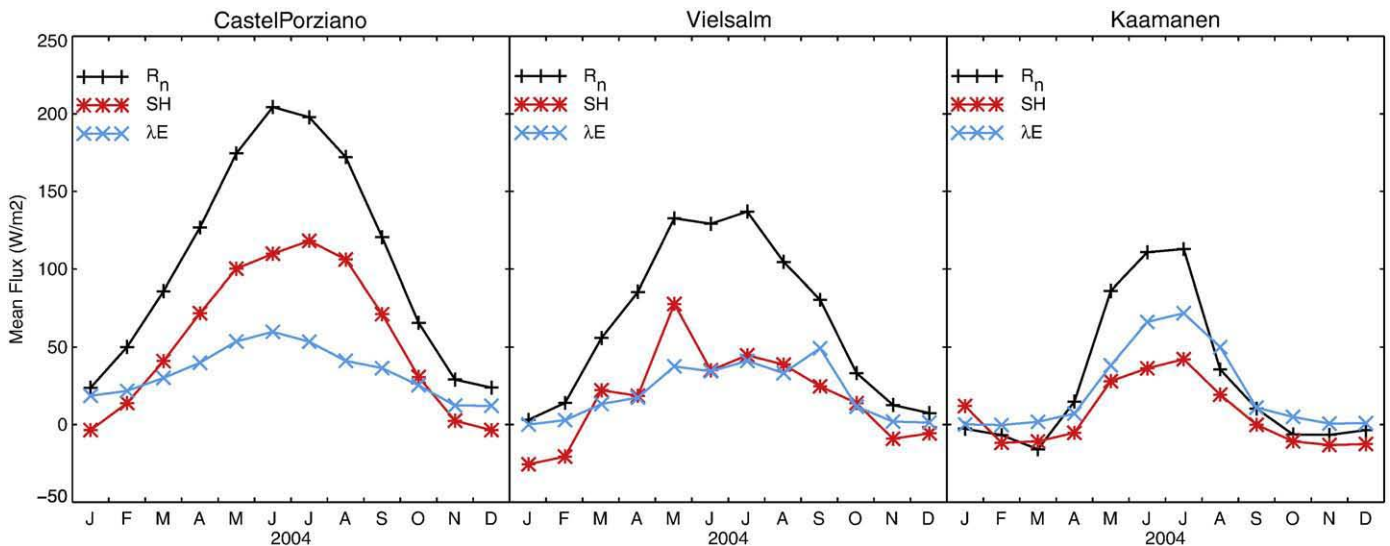


Fig. 7. Monthly net radiation (R_n), latent heat flux (λE) and sensible heat flux (SH) during 2004 at three CarboEurope flux tower sites covering a wide range of climate zones: Castelporziano, Italy (summer-dry Mediterranean evergreen forest, Reichstein et al., 2002); Vielsalm, Belgium (temperate mixed forest, Aubinet et al., 2001); and Kaamanen, Finland (Arctic tundra, Aurela et al., 2002). [Figure adapted from Seneviratne and Stöckli, 2008].

Manabe, 1969; Budyko, 1974) are still conceptually useful to understand soil moisture–climate relationships in current climate models, at least to a first approximation.

4.4. Evidence from observations

In the preceding subsection, we have discussed modelling approaches representing evapotranspiration and its dependency on soil moisture. It is of course essential to identify whether ground observations may help in validating these relationships. Though observational datasets of soil moisture and evapotranspiration are scarce and limited in space and time (see Section 8), first investigations have attempted to address this issue. In particular flux measurements from the FLUXNET project (e.g. Baldocchi et al.,

2001; Baldocchi, 2008) can be useful for identifying the processes controlling evapotranspiration on the local and regional scales.

As an example, we show in Fig. 7 measurements of net radiation, latent and sensible heat flux at three European FLUXNET sites (adapted from Seneviratne and Stöckli, 2008). The considered sites encompass Castelporziano (Italy, summer-dry Mediterranean evergreen forest; left), Vielsalm (Belgium, temperate mixed forest; middle), and Kaamanen (Finland, Arctic tundra; right). From left to right, the figure displays the gradient from the transitional Mediterranean climate regime (with soil moisture limitation) to increasingly wet and cold climate. This analysis illustrates well how drivers of evapotranspiration vary with climate regimes, with soil moisture being limiting in the Mediterranean site (left), while the low amount of evapotranspiration at the arctic tundra site is due to radiation limitation and the short growing season length (right). Interestingly,

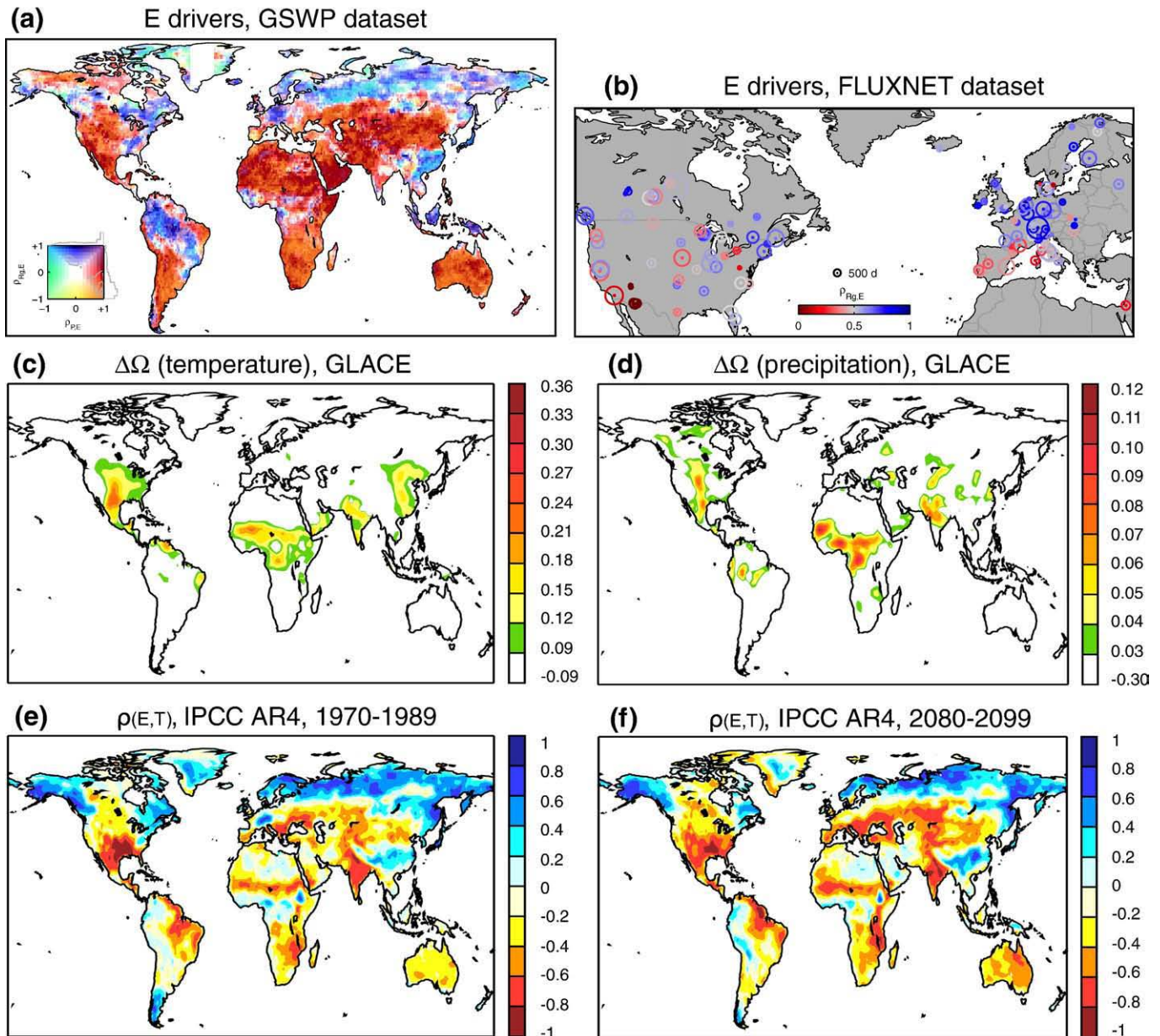


Fig. 8. Land–atmosphere coupling diagnostics based on model and observational data. Top row: Estimation of the drivers of evapotranspiration (moisture and radiation; after Teuling et al., 2009) based on simulations from the GSWP project (left, yearly correlations with global radiation R_g and precipitation P , redrawn for whole globe) and FLUXNET observations (right, daily correlations with global radiation). Middle row: Estimations of the soil moisture–temperature (left) and soil moisture–precipitation (right) coupling, based on $\Delta\Omega$ -coupling diagnostic and output of 12 GCMs from the GLACE-1 project (after Koster et al., 2004a, 2006). Bottom row: Estimations of soil moisture–temperature coupling for 1970–1989 (left panel) and 2080–2099 (right panel) climates, based on 3 IPCC GCMs and diagnosed with $\rho(E, T)$ (after Seneviratne et al., 2006a,b). See Appendix B for more details on land–atmosphere coupling diagnostics. [Figures adapted and redrawn from Koster et al., 2004a (d); Koster et al., 2006 (c); Seneviratne et al., 2006a (e,f); Teuling et al., 2009 (a,b)].

this results in similar values of summer evapotranspiration at the three sites. A more detailed analysis of evapotranspiration drivers on the continental scale for Europe and North America (Fig. 8b, from Teuling et al., 2009) confirms that regions with distinct evapotranspiration regimes (i.e. radiation-limited or not, Section 4.1 and Fig. 5) can be identified from FLUXNET measurements. Despite local variations, the location of these regions for Europe and America, diagnosed from the correlation of (daily) measured evapotranspiration with measured global radiation ($\rho_{(R_g,E)}$, Fig. 8b), roughly agrees with estimates of evapotranspiration regimes based on LSM simulations from the Global Soil Wetness Project driven with observation-based forcing (GSWP, Dirmeyer et al., 2006a, Fig. 8a; based on yearly correlations of evapotranspiration with global radiation and precipitation).

Satellite data and ground observations also point to a similar geographical distribution of limitation regimes (energy- vs. soil moisture-limited) for net primary production (Nemani et al., 2003). For instance, in humid tropical regions (e.g. Amazonia), in which forests are rarely exposed to soil moisture limitation due to ample supply and the presence of deep roots (e.g. Nepstad et al., 1994; Schenk and Jackson, 2005), vegetation activity may be higher in the dry than in the wet season, because evapotranspiration and photosynthesis are radiation-limited (cloud cover) during the latter (Saleska et al., 2007). Some regions may switch between energy-limited and soil moisture-limited evapotranspiration regimes over the course of the year (e.g. Ryu et al., 2008), and the switch between the two regimes may also be dependent on land cover (e.g. Zhang et al., 2001; Zaitchik et al., 2006). Note that in those regions showing soil moisture dry-down and subsequent decrease of evapotranspiration, an approximately linear dependency of evapotranspiration on soil moisture (Fig. 5 and Eq. (8b), see also Appendix B.3) appears to be consistent with observations (Teuling et al., 2006).

Thus, one can conclude that there is an overall qualitative agreement between modelling and observational results regarding the existence of distinct evapotranspiration (and photosynthesis) regimes, and that these can to a first-order be divided in soil moisture- and energy-limited regimes (Section 4.1 and Fig. 5). However, there is still significant uncertainty regarding the precise geographical definition of the transitional climate regions, the exact strength of coupling between soil moisture and evapotranspiration, and the potential role of other evapotranspiration drivers such as CO₂ concentrations and nutrient limitation (see further sections).

5. Soil moisture–climate interactions and feedbacks: Impacts on temperature and precipitation

5.1. Soil moisture–temperature coupling

As mentioned previously, an important impact of soil moisture on near-surface climate is related to changes in air temperature: Whenever soil moisture limits the total energy used by latent heat flux, more energy is available for sensible heating, inducing an increase of near-surface air temperature. Contrary to soil moisture–precipitation interactions (Section 5.2), soil moisture–temperature interactions have only received attention recently in the climate modelling community (e.g. Brabson et al., 2005; Koster et al., 2006; Clark et al., 2006; Seneviratne et al., 2006a). Nonetheless, they can significantly impact near-surface climate, and have been shown in particular to be relevant for the occurrence of extreme hot temperatures and heat waves (e.g. Durre et al., 2000; Seneviratne et al., 2006a; Fischer et al., 2007a,b; Diffenbaugh et al., 2007; Zhang et al., 2009; Jaeger and Seneviratne, in press).

5.1.1. Processes leading to soil moisture–temperature coupling

Fig. 9 displays the main interactions leading to soil moisture–temperature coupling and soil moisture–temperature feedbacks (for a

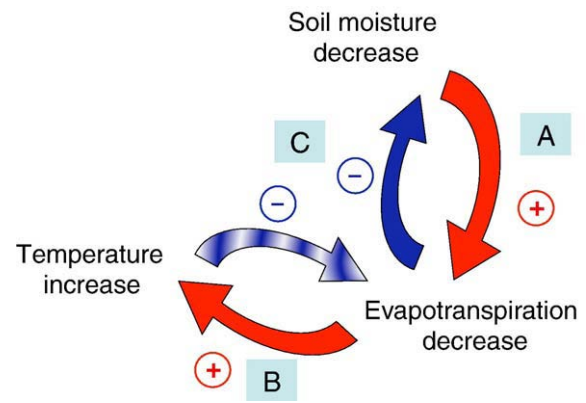


Fig. 9. Processes contributing to soil moisture–temperature coupling and feedback loop. Positive arrows (red) indicate processes that lead to a drying/warming in response to a negative soil moisture anomaly and blue arrows denote potential negative feedbacks (the hatched blue arrow indicates the tendency for enhanced temperature to lead to more evaporative demand; if this results in an increase of evapotranspiration, this in turn leads to a further drying of the soil and thus to a positive feedback loop). Note that possible links to the radiation (clouds and water vapour) are not included, and that the displayed relationships similarly apply for positive soil moisture anomalies (leading to negative temperature anomalies). (A), (B) and (C) refer to the different steps of the feedback loop (see text for details).

definition of the terms “coupling” and “feedback” see Appendix A). The displayed feedback loop indicates the possible impact of a negative soil moisture anomaly for a positive anomaly of temperature, mediated through a negative anomaly of evapotranspiration. Positive arrows (red) indicate processes that lead to a drying/warming and blue arrows denote potential negative feedbacks (the hatched blue arrow indicates the tendency for enhanced temperature to lead to more evaporative demand, while the filled blue arrow indicates the impact of evapotranspiration on soil moisture content). Note that possible links to the radiation (clouds and water vapour) are not included. Relationship (A) refers to the link between soil moisture and evapotranspiration, which has been addressed in Section 4 (but in this case indicating the impact of a negative anomaly of soil moisture for a consequent negative anomaly of evapotranspiration). As mentioned, this link is expected to be strongest in transitional zones between dry and wet climates. Relationship (B) relates to the link between evapotranspiration and sensible heat flux, the latter ultimately impacting air temperature: Decreased evapotranspiration leads to an increase in sensible heat flux and thus an increase in air temperature. Relationship (C) relates to a potential positive feedback (induced by a chain of two negative feedbacks) leading to a further temperature increase: Increased temperature leads to a higher vapour pressure deficit and evaporative demand, and thus to a potential increase in evapotranspiration despite the dry conditions, possibly leading to a further decrease in soil moisture. The feedback loop can go on until the total drying of the soil, when temperature increases cannot be damped by any further increases in evapotranspiration.

5.1.2. Regions of strong soil moisture–temperature coupling

Fig. 8c,e displays two global estimates of the strength of soil moisture–temperature coupling in boreal summer (June–July–August, JJA) based on modelling datasets, in one case from the GLACE experiment (Fig. 8c; Koster et al., 2006) and in the other case from IPCC AR4 simulations (Fig. 8e; Seneviratne et al., 2006a). The two employed coupling diagnostics are introduced in the Appendix B. The GLACE estimate corresponds to the ΔQ diagnostic (Eq. (18), see Appendix B.1) applied to 3-month long ensemble simulations performed for JJA 1994 sea surface temperature (SST) conditions (mean of 12 AGCMs). On the other hand, the IPCC estimate is derived from the correlation between JJA evapotranspiration and temperature ($\rho_{(E,T)}$, see Appendix B.2) over a twenty-year period (1970–1989), for the ECHAM5, GFDL, and HadGEM1 Atmospheric–Oceanic General

Circulation Models (AOGCMs). These three AOGCMs have been shown to have the most accurate representation of present-day North-hemispheric circulation patterns (van Ulden and van Oldenborgh, 2006).

Both model-based coupling estimates in Fig. 8c and e show similar patterns regarding regions of strong soil moisture–temperature coupling. Using the correlation diagnostic $\rho_{(E,T)}$ (Fig. 8e), regions with negative correlation (red) appear as coherent geographical patterns. These regions do include the hot spots of strong soil moisture–atmosphere coupling identified in the GLACE study (Fig. 8c), and fit well the definition of transitional climate zones. However, they include some regions not identified in the GLACE study, in particular the Mediterranean climate zone (but there are also differences in e.g. equatorial Africa). This may be due to the fact that the GLACE study was performed using the SST conditions of only one year (1994), compared to the 20-year period considered in the IPCC AR4 analysis (Seneviratne et al., 2006a). Moreover, the GLACE experiment focused on impacts of soil moisture for subseasonal climate variability, while the $\rho_{(E,T)}$ diagnostic was applied to the IPCC simulations to investigate the role of soil moisture for interannual variations in temperature. Note that the GCM-based hot spots identified in Fig. 8c and e for present climate appear overall consistent with the geographical distribution of moisture-limited evapotranspiration regimes diagnosed from the GSWP-2 dataset (Fig. 8a; these latter regions are more extended since they also include dry climate regimes in addition to transitional climate regimes).

5.1.3. Evidence from observations

The two estimates of soil moisture–temperature coupling (c and e) displayed in Fig. 8 are based on modelling datasets. However, the comparison of the derived patterns with the correlation of measured evapotranspiration with global radiation as derived from FLUXNET observations (Fig. 8b), suggests that the regions with soil moisture-limited regime (i.e. low $\rho_{(Rg,E)}$) are overall correctly captured in the models (with the exception of the Mediterranean region for the GLACE simulations), though the observations show strong spatial variability in North America. This is also consistent with the recent results of Koster et al. (2009a), who investigated the links between regional precipitation availability and interannual temperature variability from observational datasets to derive indirect observations-based estimates of soil moisture–temperature coupling: Overall, the inferred patterns are similar to those in Fig. 8c,e (though not perfectly overlapping), and a signal is identified in the Mediterranean region as suggested by the $\rho_{(E,T)}$ analysis of the IPCC AR4 simulations (Fig. 8e).

5.2. Soil moisture–precipitation coupling

5.2.1. Historical overview

The investigation of soil moisture–precipitation coupling has been the subject of much research for several years to decades. First speculations on the impact of soil moisture for precipitation date back to the 19th century (see e.g. Holzman, 1937, cited in Eltahir, 1998). In the second half of the 20th century, early modelling studies investigated the potential role of soil moisture for global and regional climate with sensitivity experiments (e.g. Charney, 1977; Shukla and Mintz, 1982; Yeh et al., 1984). These investigations spurred a long-lasting interest in this topic.

Until the 1990s, many studies focused on the concept of moisture recycling, i.e. the quantification of the ratio of precipitated water that is directly contributed by regional evapotranspiration (see e.g. Brubaker et al., 1993, and Eltahir and Bras, 1996), with possible inferences regarding corresponding land–atmosphere interactions (e.g. Rodríguez-Iturbe et al., 1991; Entekhabi et al., 1992). Most of these analyses were based on simple budget calculations using either model or observational data (e.g. Budyko, 1974; Eltahir and Bras, 1994). More detailed approaches using water-vapour tagging in

climate models or the computation of back-trajectories have also been successfully implemented (e.g. Joussaume et al., 1984; Koster et al., 1986; Dirmeyer and Brubaker, 1999; Brubaker et al., 2001; Bosilovich and Schubert, 2002; Bosilovich and Chern, 2006).

Recent studies on soil moisture–precipitation coupling have, however, emphasized more prominently the role of indirect interactions, rather than moisture recycling per se (e.g. Ek and Mahrt, 1994; Betts et al., 1996; Beljaars et al., 1996; Paegle et al., 1996; Giorgi et al., 1996; Eltahir, 1998; Schär et al., 1999; Pal and Eltahir, 2008; Findell and Eltahir, 2003a,b; Ek and Holtslag, 2004; Betts, 2004; Taylor and Ellis, 2006; Taylor et al., 2007; Alfieri et al., 2008; Hohenegger et al., 2009; Santanello et al., 2009). These mechanistic studies highlight that in many instances the key for understanding soil moisture–precipitation interactions lies more in the impact of soil moisture anomalies on boundary-layer stability and precipitation formation than in the absolute moisture input resulting from modified evapotranspiration. For instance, the additional precipitated water falling over wet soils may originate from oceanic sources, but the triggering of precipitation may itself be the result of enhanced instability induced by the wet soil conditions (or in some cases dry soil conditions, see Section 5.2.2). Moreover, also non-local feedbacks can be important, i.e. advection of evaporated moisture for neighbouring land regions (e.g. Rowell and Blondin, 1990; Beljaars et al., 1996; see also Section 6.2). In addition, several studies have also highlighted the possible role of land surface heterogeneity for the generation of mesoscale features and resulting precipitation formation (e.g. Chen and Avissar, 1994; Pielke et al., 1998). In the next subsection, we discuss in more detail the chain of processes playing a role in soil moisture–precipitation feedbacks.

5.2.2. Processes leading to soil moisture–precipitation coupling and soil moisture–precipitation feedback

As a conceptual framework, we depict in Fig. 10 a simplified representation of the processes that contribute to soil moisture–precipitation coupling and soil moisture–precipitation feedback: (A) the relationship between soil moisture anomalies and subsequent evapotranspiration anomalies; (B) the relationship between evapotranspiration anomalies and subsequent precipitation anomalies; (C) the relationship between precipitation and subsequent soil moisture anomalies. Soil moisture–precipitation coupling refers to steps (A) and (B), while soil moisture–precipitation feedback refers to the full cycle (A–C)—see also Appendix A. We will discuss here these three steps (A–C), starting with relationship (C), which is the easiest to establish, and closing with relationship (B), which is the most subject to uncertainty.

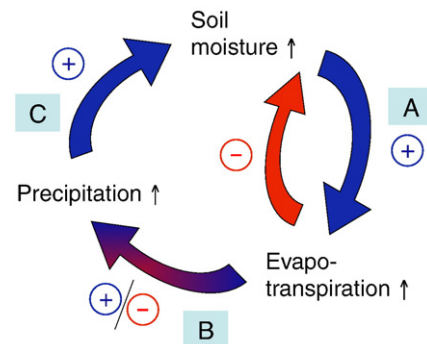


Fig. 10. Processes contributing to soil moisture–precipitation coupling and feedback loop. Positive arrows (blue) indicate processes leading to a positive soil moisture–precipitation feedback (wetting for positive soil moisture anomaly, drying for negative soil moisture anomaly), the negative arrow (red) indicates a potential negative feedback damping the original soil moisture anomaly, and the red-blue arrow indicates the existence of both positive and negative feedbacks between evapotranspiration and precipitation anomalies. (A), (B) and (C) refer to the different steps of the feedback loop (see text for details).

Relationship (C) (higher precipitation leading to higher soil moisture) can generally be considered trivial, though there are a few exceptions. For instance in case of precipitation over very wet or saturated soils, for which anomalies in precipitation will result in anomalies of runoff rather than soil moisture. Similarly, in cases of intense precipitation, when the precipitation rate is higher than the infiltration rate, and thus precipitation anomalies again result in runoff rather than soil moisture anomalies (Horton vs Dunne runoff generation: see Horton, 1933; Dunne, 1978; Bras, 1990). Conversely, under extreme drought, soils may present cracks in which precipitation infiltrates, bypassing the surface zone (e.g. Hillel, 1998). Under certain conditions, air may also be so dry that the rain evaporates before reaching the surface, an effect that may be of importance in arid regions (Small et al., 1999; Pal et al., 2000). Nonetheless, except for these few extreme cases, the positive (C) relationship can be assumed to hold.

Relationship (A) (higher soil moisture leading to higher evapotranspiration) refers to the interaction between soil moisture and evapotranspiration (Section 4). As mentioned earlier, the double constraint of having an impact of soil moisture on evapotranspiration, and this impact to be noticeable in terms of resulting evapotranspiration, means that one can expect a positive and significant relationship (A) only in transitional regions between dry and wet climates (Sections 4.1 and 4.3, and Figs. 5 and 6). In terms of soil moisture–precipitation feedbacks, as indicated with the red arrow, a negative feedback potentially exists at this level, since increasing evapotranspiration decreases the available soil moisture. Thus, in order to sustain a positive soil moisture–precipitation feedback, it is not sufficient that enhanced soil moisture and evapotranspiration lead to additional precipitation: The enhancement in precipitation needs to be at least equivalent to the enhancement in evapotranspiration, else the net effect will be a reduction of the original soil moisture anomaly.

Relationship (B) (higher evapotranspiration leading to higher precipitation) is the most uncertain link in our current understanding of the soil moisture–precipitation coupling/feedback, as well as the most difficult to ascertain given the number of processes involved. Many modelling studies have investigated this issue. A large number of them identified a possible positive relationship between enhanced soil moisture and evapotranspiration on the one side and precipitation on the other side (Shukla and Mintz, 1982; Yeh et al., 1984; Trenberth et al., 1988; Namias, 1991; Milly and Dunne, 1994; Betts et al., 1996; Beljaars et al., 1996; Trenberth and Guillemot, 1996; Schär et al., 1999; Koster et al., 2000; Pal and Eltahir, 2008; Betts, 2004; Schubert et al., 2004; Koster et al., 2004a, 2006). However, the involved processes are complex, and several further modelling studies suggested that under specific conditions convective instability and/or cloud formation may be stronger over dry soils (e.g. Ek and Mahrt, 1994; Paegle et al., 1996; Giorgi et al., 1996; Findell and Eltahir, 2003a,b; Ek and Holtslag, 2004; Hohenegger et al., 2009). The latter results appear up-to-now mostly confined to single-column, cloud-resolving, and some regional climate simulations, and may also depend on the choice of the convective parameterizations (e.g. Giorgi et al., 1996). Moreover, effects of land surface heterogeneity for the inferred feedbacks also need to be taken into account (Chen and Avissar, 1994; Pielke et al., 1998). In general, current global climate models do, however, not present significant negative feedbacks between soil moisture and precipitation (Koster et al., 2004a), though they do not necessarily agree on the magnitude of the soil moisture–precipitation coupling strength (Koster et al., 2004a; Lawrence and Slingo, 2005; Koster et al., 2006; Guo et al., 2006a; see also Section 5.2.3). One exception is the study by Cook et al. (2006), with the Community Climate System Model version 3, which did report potential negative soil moisture–precipitation feedbacks in South Africa.

Some observational studies also investigated the possible relationship between enhanced soil moisture/evapotranspiration and

precipitation, but the results have been inconclusive up to now, with some indications of positive, negative, and no feedbacks (see Section 5.2.4). Thus, as mentioned at the beginning of this subsection, relationship (B) can be seen as the crucial step in the soil moisture–precipitation feedback loop, and further investigations will be necessary to fully characterize the links between evapotranspiration and precipitation.

5.2.3. Regions of strong soil moisture–precipitation coupling

In a modelling framework, the GLACE investigation has characterized the strength of coupling between soil moisture and precipitation in current state-of-the-art AGCMs (Koster et al., 2004a). The resulting map (Fig. 8d) is very similar to the map regarding the strength of coupling between soil moisture and temperature in the same AGCMs (Fig. 8c). This suggests that in these models, the same process controls both couplings, namely the evapotranspiration sensitivity to soil moisture (Section 4; relationship (A) in the preceding subsection). However, it is important to note the range of behaviour between models (Koster et al., 2004a, 2006; Guo et al., 2006a). For instance, some of the participating models display very little or no coupling (Lawrence and Slingo, 2005).

As mentioned in the preceding subsection, the GLACE AGCMs may not be able to represent negative feedbacks between evapotranspiration and precipitation, which are found in some single-column, cloud-resolving and regional climate simulations and the AGCM study of Cook et al. (2006). Investigations by Findell and Eltahir (2003a,b) have for instance identified areas with negative, positive, and no soil moisture–precipitation feedbacks in the continental United States, where AGCMs consistently only display a positive feedback (Koster et al., 2004a, 2006). This highlights the need to identify regions of strong soil moisture–precipitation coupling not only based on AGCMs studies, but also using other modelling tools, and most importantly observational datasets (despite associated methodological issues, see Section 5.2.4).

5.2.4. Evidence from observations

In the view of the large discrepancies between modelling studies (see preceding subsections), it appears increasingly necessary to investigate soil moisture–precipitation relationships with observational data. However, the scarcity of the soil moisture measurements, combined with the low temporal resolution of most available long-term datasets, is a critical issue (Section 8). Moreover, because of the direct impact of precipitation on soil moisture, links of causality between soil moisture and precipitation are difficult to establish (e.g. Salvucci et al., 2002; Wei et al., 2008a; Orlowsky and Seneviratne, *in press*; see also Appendix B.2).

Accordingly, direct analyses of soil moisture–precipitation feedbacks using soil moisture and precipitation observations have been inconclusive so far (e.g. Findell and Eltahir, 1997; Salvucci et al., 2002; Kochendorfer and Ramirez, 2005; Alfieri et al., 2008). Moreover, some features of observational soil moisture datasets such as bimodality, which could potentially be compatible with positive soil moisture–precipitation feedbacks (e.g. D'Odorico and Porporato, 2004; D'Andrea et al., 2006), can also be explained independently of such interactions (Teuling et al., 2005). Using precipitation data over the United States, Koster et al. (2003) found that the observed persistence patterns could only be reproduced with the NSIPP AGCM in simulations with interactive soil moisture. However, the patterns in the model were not in full agreement with the observations, and it is possible that other conclusions could be reached with another AGCM, given the range of soil moisture–precipitation coupling in current climate models.

Future studies on soil moisture–precipitation feedbacks will need to take into account the methodological issues underlying the establishment of causal links between soil moisture anomalies and resulting precipitation anomalies, given the potentially confounding

effects of the impact of precipitation on soil moisture, and that of other drivers inducing persistence in precipitation (e.g. [Orlowsky and Seneviratne, in press](#)). One approach is to investigate single aspects of the feedback loop in isolation, for instance the relationships (A) and (B) above (impact of soil moisture on evapotranspiration, and impact of evapotranspiration on precipitation, respectively). This will also require more comprehensive observational datasets, for further regions, longer time periods, and/or with higher temporal and spatial resolution, in order to investigate the underlying processes in the necessary detail. These would also allow a better evaluation of the respective relationships and processes in current climate models. We will discuss perspectives on soil moisture and evapotranspiration validation datasets in [Section 8](#).

6. Soil moisture–climate interactions and feedbacks: Further relevant aspects

6.1. Climate persistence associated with soil moisture

Another aspect of importance in the context of soil moisture–climate interactions is the persistence associated with soil moisture anomalies. Similar to the oceans, which can store heat and thus induce persistence (“memory”) in the climate system, land represents a water storage for the climate system, with associated persistence features. Moreover, since evapotranspiration (latent heat flux) from land is both a water and energy flux ([Section 3](#)), soil moisture is both a water and energy storage. This also applies to other forms of water storage on land such as snow, surface water, and in some cases groundwater.

The associated “soil moisture memory” is an important component in land–atmosphere interactions, and has thus been the topic of numerous observational and modelling investigations (e.g. [Vinnikov et al., 1996](#); [Liu and Avissar, 1999](#); [Entin et al., 2000](#); [Koster and Suarez, 2001](#); [Schlosser and Milly, 2002](#); [Wu and Dickinson, 2004](#); [Seneviratne et al., 2006b](#); [Lorenz et al., in press](#)). In particular, several studies have addressed the potential use of soil moisture initialization for subseasonal and seasonal forecasting (e.g. [Fennessy and Shukla, 1999](#); [Koster et al., 2000](#); [Koster et al., 2004a,b](#); [Ferranti and Viterbo, 2006](#); [Kanae et al., 2006](#); [Conil et al., 2007](#); [Koster et al., 2010](#)). Note that the potential of soil moisture for such applications depends both on the strength of soil moisture–climate coupling and the respective soil moisture memory in the given regions. The resulting “hot spots” of soil moisture initialization for seasonal forecasting may look slightly different from hot spots of land–atmosphere coupling, due to the impact of soil moisture memory ([Seneviratne et al., 2006b](#)).

6.2. Large-scale and non-local impacts of soil moisture anomalies

While it is well possible that soil moisture can have large-scale or non-local impacts (e.g. by impacting large-scale circulation patterns), only few studies have investigated this topic up-to-now. In the early modelling study of [Shukla and Mintz \(1982\)](#), based on extreme sensitivity experiments with a General Circulation Model (GCM), a forced decrease of evapotranspiration was found to induce strong thermal lows with some impacts on the monsoon circulation in India (dry conditions inducing a strengthened monsoon). A later study by [Meehl \(1994\)](#) pointed out, however, that there can be competing positive and negative feedbacks between soil moisture and monsoon precipitation. [Douville \(2002\)](#) also identified significant impacts of soil moisture on the African monsoon, with possible remote effects of soil moisture anomalies on circulation patterns. Note that in the Sahel region, drought has also been reported as a possible factor leading to the buildup of dust, which in turn may affect the radiation balance and circulation patterns ([Nicholson, 2000](#)).

Further modelling studies investigated possible impacts of soil moisture on the North American Monsoon System (NAMS): [Beljaars](#)

[et al. \(1996\)](#) found for instance improved forecasting skill in the central United States with modified evapotranspiration conditions 1 day upstream. Furthermore, [Small \(2001\)](#) suggested that some of the inferred feedbacks in this region may depend on the exact location of the soil moisture anomalies, and [Kanamitsu and Mo \(2003\)](#) proposed that enhanced evaporation over Arizona and New Mexico could lead to a stronger moisture flux from the Gulf of Mexico.

It has also been suggested that soil moisture anomalies in extratropical regions can produce large-scale and/or non-local effects, although these are not necessarily associated with large-scale circulation changes. In a pioneering modelling study, [Rowntree and Bolton \(1983\)](#) have investigated the effect of soil moisture anomalies over Central Europe in a global atmospheric model. They found that such anomalies affect temperature, humidity and rainfall by altering the partitioning of turbulent fluxes. An interesting aspect of their results is that the effect on precipitation is not limited to the region experiencing soil moisture anomalies but also extends north of the anomaly area into southern Scandinavia. This propagation of the anomaly arises from the advection of drier (or wetter) air by southerly episodes. [Vautard et al. \(2007\)](#) identify a similar mechanism to explain the occurrence of extreme heat waves in Europe. By analysing meteorological records, they suggest that winter drought in Southern Europe is a necessary condition for the development of major summertime heat waves in Europe, with advection of dry air from Southern Europe contributing to the propagation of the heat wave to Northern Europe. Note that in these studies the highlighted large-scale effect of soil moisture does not involve changes in circulation patterns but rather rely on modification of the air properties (humidity for instance), which are in turn advected by the large-scale circulation. Nonetheless, both in global ([Haarsma et al., 2009](#)) and regional climate simulations ([Pal and Eltahir, 2003](#); [Fischer et al., 2007a](#)), possible effects of soil moisture on pressure systems during droughts were also recently documented. In addition, the GCM study by [Cook et al. \(2006\)](#) for South Africa identified possible negative feedbacks of soil moisture on precipitation due, among other factors, to induced anomalous surface divergence and increased subsidence. At present the evidence regarding impacts of soil moisture on extratropical circulation patterns is, however, still scarce and based on single-model studies.

6.3. Soil moisture–albedo interactions

The possible role of (land) albedo feedbacks has been the subject of several investigations since the pioneering study by [Charney \(1977\)](#). The relevant mechanisms are associated with a three-fold impact of soil moisture on land surface albedo: 1) changes in bare soil albedo with soil moisture content; 2) changes in vegetation albedo with soil moisture content; 3) changes in exposed bare soil fraction due to drought impacts on vegetation (leaves' orientation, leaf area index). As reported in recent studies, the effects of droughts on surface albedo may actually be smaller than previously assumed (e.g. [Eltahir, 1998](#)) because of contrasting changes in the visible and near-infrared spectral bands (e.g. [Teuling and Seneviratne, 2008](#)). These effects may also depend on the region of study ([Zaitchik et al., 2007](#)).

6.4. Soil moisture interactions with biogeochemical cycles

Because soil moisture impacts plants' productivity and survival, it also has strong impacts on the biogeochemical cycles. For instance, the assimilation of CO₂ in plants is coupled with their water loss through transpiration (e.g. [Farquhar et al., 1980](#); [Farquhar and Sharkey, 1982](#); [Ball et al., 1987](#); [Field et al., 1995](#); [Sellers et al., 1997](#)). While it is not clear at present what would be the exact impact of enhanced CO₂ on plants' photosynthesis (e.g. [Cox et al., 2000](#); [Führer, 2003](#); [Körner et al., 2005](#); [Norby et al., 2005](#); [Friedlingstein](#)

et al., 2006; Leuzinger and Körner, 2007; Finzi et al., 2007; Thornton et al., 2007; Sokolov et al., 2008; Bonan, 2008) and ultimately on the water cycle (e.g. Gedney et al., 2006; Piao et al., 2007; Betts et al., 2007), there is no doubt that changes in the water cycle for their part would have impacts on the terrestrial carbon exchange through this coupling. As an example, during the extreme drought and heat wave summer of 2003 in Europe, it was found that plants became less productive and, as a consequence, the European continent became a source of CO₂ in summer (Ciais et al., 2005; Granier et al., 2007; Reichstein et al., 2007). This possibly even had an impact on the measured CO₂ record at Mauna Loa (Jones and Cox, 2005). Also the 2005 drought in the Amazon rainforest had a major impact on carbon uptake in this region (Phillips et al., 2009). Accordingly, recent studies suggest that currently occurring changes in soil moisture distribution are already impacting the carbon cycle (e.g. Angert et al., 2005; Canadell et al., 2007). Other biogeochemical cycles influenced by vegetation and terrestrial water fluxes should similarly be affected by, or impact on, variations in soil moisture content, especially in the context of climate change (be it indirectly through links to the carbon cycle). This is in particular the case for the nitrogen cycle (e.g. Hungate et al., 2003; Finzi et al., 2007; Thornton and Zimmermann, 2007; Thornton et al., 2007; Sokolov et al., 2008; Gruber and Galloway, 2008).

6.5. Role of land cover and vegetation dynamics for soil moisture–climate interactions

Soil moisture impact on evapotranspiration is mediated by transpiration through plants for the most part (Dirmeyer et al., 2006a). Hence, vegetation characteristics (land cover) and dynamics (on various time scales) are essential for these interactions. Furthermore, effects of land cover changes or vegetation dynamics for climate are often associated with soil moisture modifications, and shifts in vegetation cover themselves may be due to shifts in soil moisture regimes.

For instance, in a recent study for the European 2003 heat wave, Zaitchik et al. (2006) have shown how geographical variations in land cover types (forests vs. pastures and crops) for a dry area in France were clearly reflected in local temperature anomalies due to differences in adaptation to drought and heat stress. Similarly, Granier et al. (2007) also identified that the sensitivity to drought in forest stands during the European 2003 heat wave was ecosystem specific. Reversely, several (model-based) projected impacts of land cover and land use changes operate through changes in soil moisture and the Bowen ratio ($\lambda E/SH$), and consequent impacts on climate (e.g. Nobre et al., 1991; Xue, 1996; Claussen et al., 2004; Foley et al., 2004; Feddesma et al., 2005; Davin et al., 2007; Bonan, 2008; Pitman et al., 2009). This also includes possible changes associated with enhanced irrigation (e.g. Boucher et al., 2004). Major uncertainties remain in the representation of physical vegetation properties, which can result in additional model divergences in projections of land cover and land use changes scenarios (Pitman et al., 2009).

For similar reasons, also vegetation dynamics can be of relevance for soil moisture–climate feedbacks, and at the same time be influenced by the latter (e.g. Prentice et al., 1992; Claussen et al., 2004; Rietkerk et al., 2004), possibly inducing multiple equilibria and critical transitions within the climate system on longer time scales (Claussen, 1998; Brovkin et al., 1998; Wang and Eltahir, 2000; Scheffer et al., 2005). Given the large variations in plants' and ecosystems' characteristics, e.g. regarding albedo, roughness length, as well as water, energy and biogeochemical exchanges, soil moisture induced shifts in vegetation cover can lead to a range of additional feedbacks on decadal to centennial scales. This is also relevant in the context of climate change (see next section).

Another aspect that is of relevance is the role of interactions between phenology and soil moisture. For instance, an early start of

the growing season may lead to enhanced spring soil moisture depletion and possible consequent impacts on climate during the summer season. There are some indications that such an effect played a possibly important role for the European 2003 heat wave (Fischer et al., 2007a; Teuling and Seneviratne, 2008). But, reversely, soil moisture stress may also impact the length of the growing season. Recent studies have illustrated that there is significant interannual variability in phenology (e.g. Asner et al., 2000; Stöckli and Vidale, 2004; Studer et al., 2005; see also Seneviratne and Stöckli, 2008), as well long-term trends thereof induced by increasing greenhouse gas concentrations (e.g. Walther et al., 2002; Parmesan and Yohe, 2003; Menzel et al., 2006; Parmesan, 2007).

7. Soil moisture–climate interactions in the context of climate change

Climate is continuously changing under the influence of natural and anthropogenic forcings. Since soil moisture–climate interactions involve a large number of processes, many of which are significantly modified in the context of climate change, one can expect that inferences based on e.g. 1960–1990 climate conditions will not hold under enhanced greenhouse gas concentrations. For instance, shifts in climate regimes modify the location of transitional climate zones, and consequently of hot spots of land–atmosphere coupling (Seneviratne et al., 2006a). Here we present briefly the projected modifications in the land water cycle and in particular in soil moisture content, together with their implications for soil moisture–climate interactions and projected climate change.

7.1. Projected changes in soil moisture and land climate

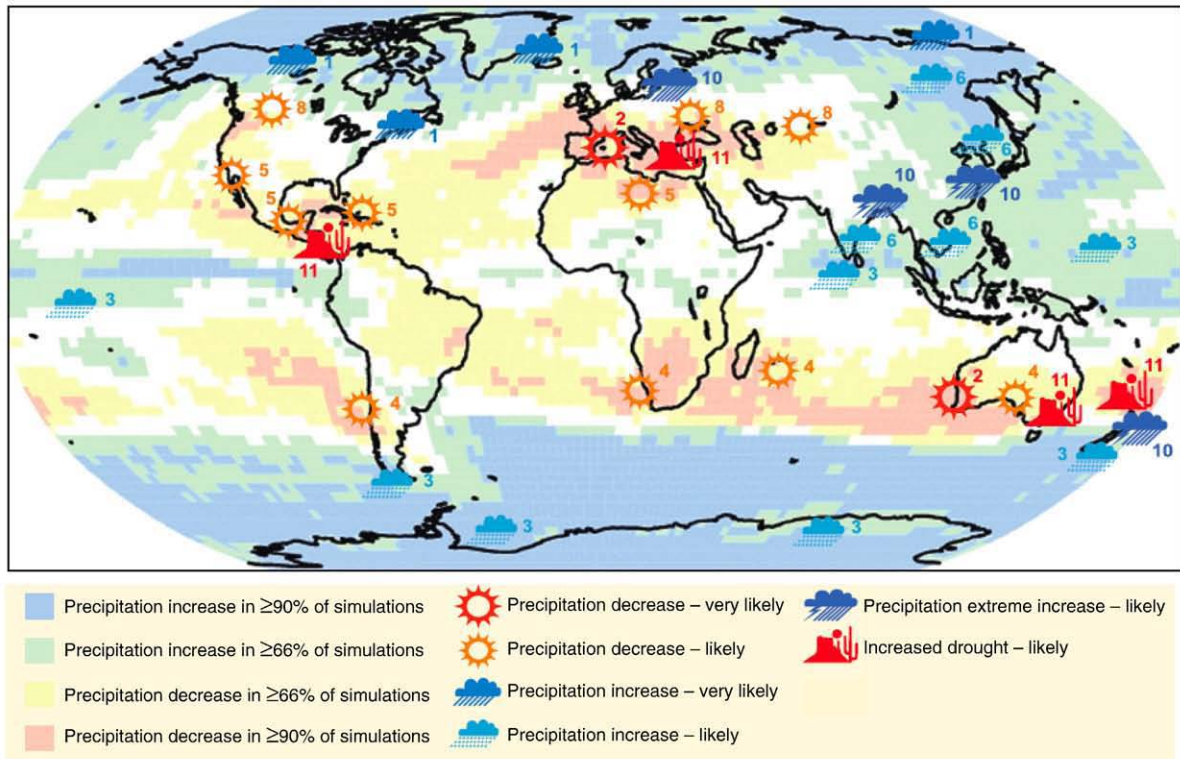
A summary on robust findings from the latest IPCC AR4 report on precipitation changes for boreal summer (JJA) is provided in Fig. 11 (top) (Christensen et al., 2007). As can be seen on this figure, climate model simulations robustly predict a decrease of mean precipitation in mid-latitude and subarid regions (especially in the Mediterranean region), and an increase in mean precipitation in the high latitudes. Together with the projected changes in radiation and temperature (which affect evapotranspiration), this strongly impacts soil moisture in current transitional regions and neighbouring areas, in particular on the European continent (Fig. 11, bottom; see also Meehl et al., 2007, and Wang, 2005, for more details).

To further illustrate the associated changes in land climate, we display in Fig. 12 (adapted from Seneviratne et al., 2006a, suppl. info) the projected changes in mean and standard deviation of boreal summer (JJA) soil moisture, evapotranspiration, precipitation, and temperature, between the time periods 1970–1989 and 2080–2099, as projected by the ECHAM5, GFDL and HadGEM1 models as part of the IPCC AR4 (A2 vs 20C3M scenarios). As mentioned, these three GCMs have been shown to have the most accurate representation of present-day North-hemispheric circulation patterns (van Ulden and van Oldenborgh, 2006).

We see a strong mean JJA soil moisture decrease in Europe as well as in the Southern U.S. under the future-climate conditions (if we exclude the high-latitude regions exhibiting very high soil moisture decreases due to decreases in snow melt). For the most part, this is due to the decreases in mean JJA precipitation (Fig. 12e) in both regions, especially in Central Europe, but is also the effect of accumulated precipitation deficits and (in some regions) evapotranspiration excess in the preceding months. The decrease in soil moisture also impacts a decrease in JJA evapotranspiration in the Southern U.S. and in the Mediterranean region, but not in Central Europe. Mean temperature changes in Europe follow a North–South gradient, with the highest changes in the Mediterranean region.

The changes in variability of the analysed variables display markedly different patterns than the changes in the mean (Fig. 12,

IPCC AR4: Robust projections wrt. changes in JJA precipitation (based on GCMs, RCMs, stat. downscaling, and process understanding)



IPCC AR4: Multi-model mean change in soil moisture (annual means, A1B vs. 20c3m, [2080-2099]-[1980-1999])

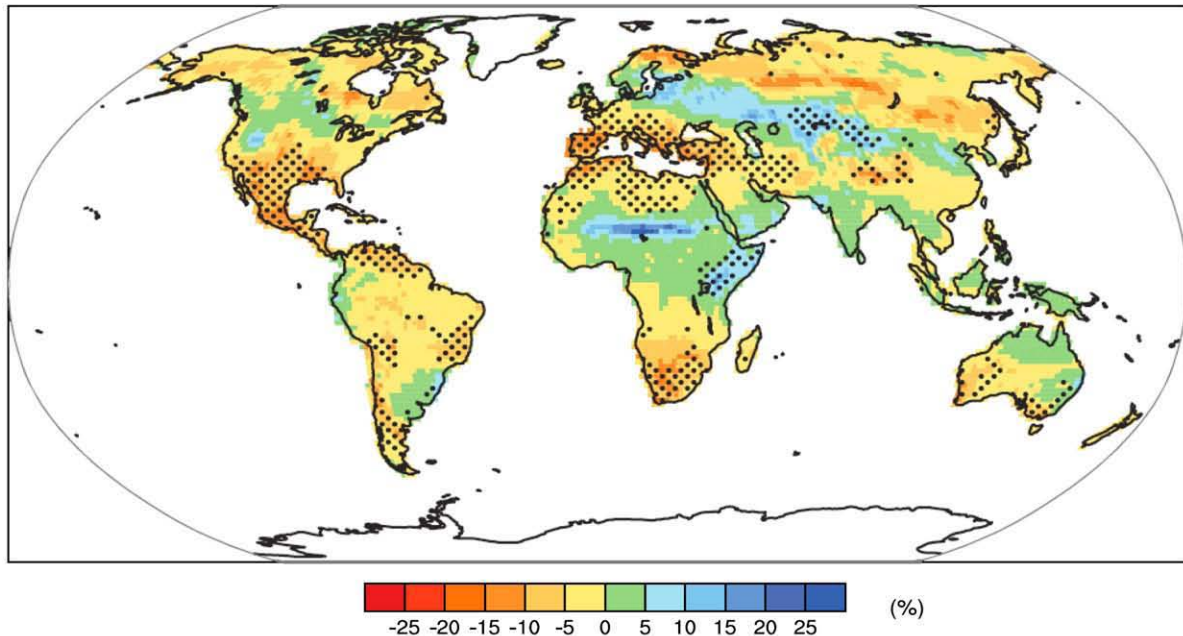


Fig. 11. (Top) Robust findings from IPCC AR4 on regional climate change for mean and extreme precipitation, and drought, for JJA. This regional assessment is based upon AOGCM and RCM simulations, statistical downscaling and process understanding. The background map indicates the degree of consistency between AR4 AOGCM simulations (21 simulations used) in the direction of simulated precipitation change. (Bottom) Projected multi-model annual mean changes in soil moisture from IPCC AR4 report [%]. Regions are stippled where at least 80% of models agree on the sign of the mean change. Changes are annual means for the SRES A1B scenario for the period 2080 to 2099 relative to 1980 to 1999, and shown at land points with valid data from at least 10 models. [Figures adapted from Christensen et al., 2007 (top) and Meehl et al., 2007 (bottom)].

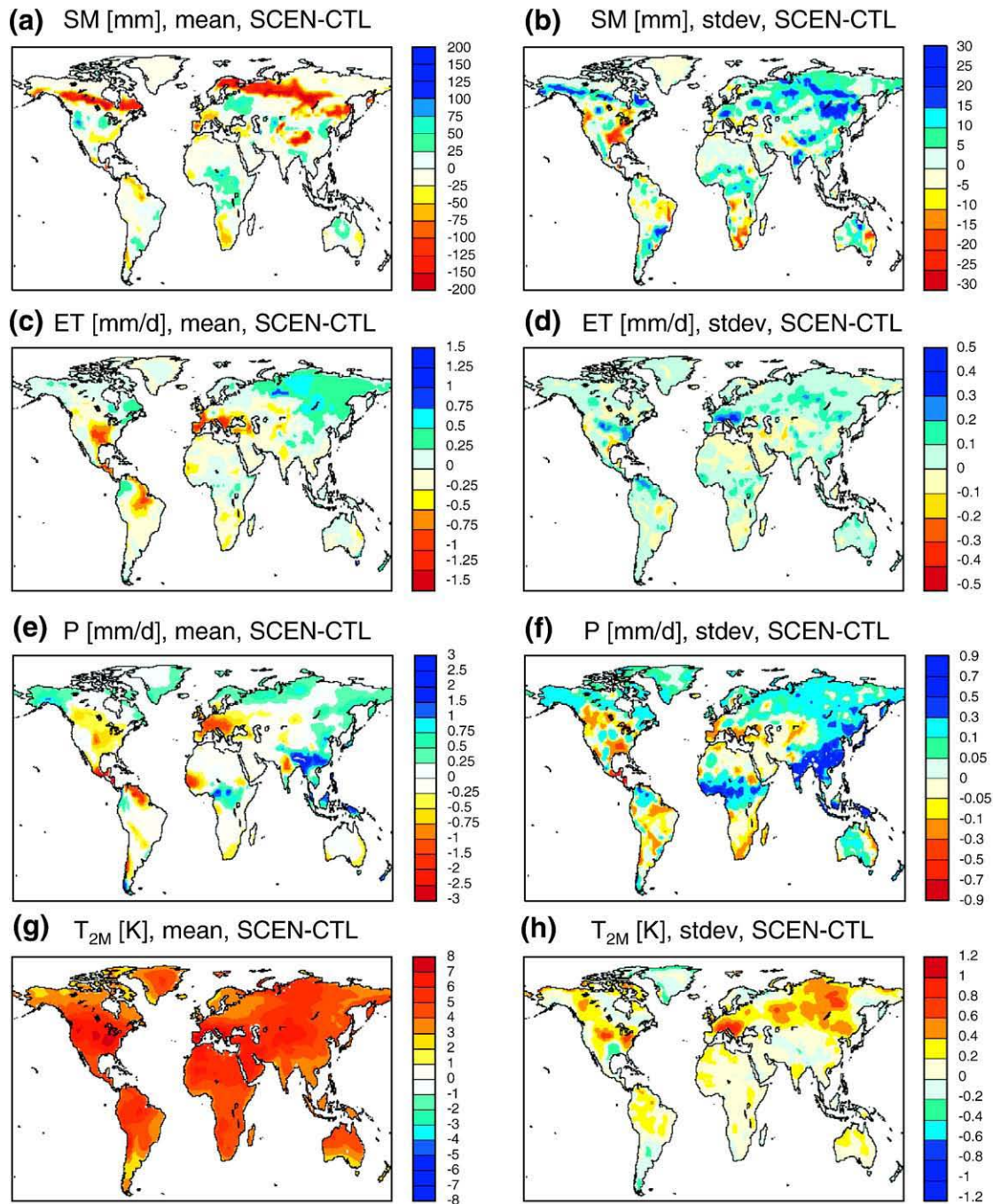


Fig. 12. Projected JJA changes in climate variables (mean, left; and standard deviation, right) derived from IPCC AR4 simulations (mean of ECHAM5, GFDL and HadGEM1), climate-change scenario (SRES A2 scenario, 2080–2099)–control (20C3M, 1970–1989): (a,b) soil moisture; (c,d) evapotranspiration; (e,f) precipitation; (g,h) 2-m temperature. [Figure adapted from Seneviratne et al., 2006a, supplementary information].

right vs. left column). The large increase in temperature variability centered over the European continent, reported in several studies (e.g. Meehl and Tebaldi, 2004; Schär et al., 2004; see also Meehl et al., 2007), is striking in this respect. Moreover, one should note that in Central Europe, increases in interannual summer variability are found for all analysed variables: temperature, but also soil moisture, evapotranspiration and precipitation. The finding that the patterns of changes in interannual climate variability are different from those in mean climate suggests that changes in extremes and climate variability cannot be directly derived from changes in mean climate, and are controlled by other mechanisms. This was indeed demonstrated for e.g. precipitation (Frei et al., 2006). As was shown by Seneviratne et al. (2006a), land–atmosphere interactions are

typically responsible for some of these divergent changes in mean climate vs. climate variability. They can also explain some inter-model difference regarding these projections (e.g. Boé and Terray, 2008). We will discuss the reasons for this in the following subsection.

7.2. Role of soil moisture changes for changes in summer climate variability

To illustrate how mean changes in soil moisture can impact summer climate variability under different climate regimes, we display in Fig. 13 (left) the hypothetical seasonal cycles of soil moisture for present (CTL) and future (SCEN) climate for a region experiencing a mean decrease in soil moisture content similar to that projected in

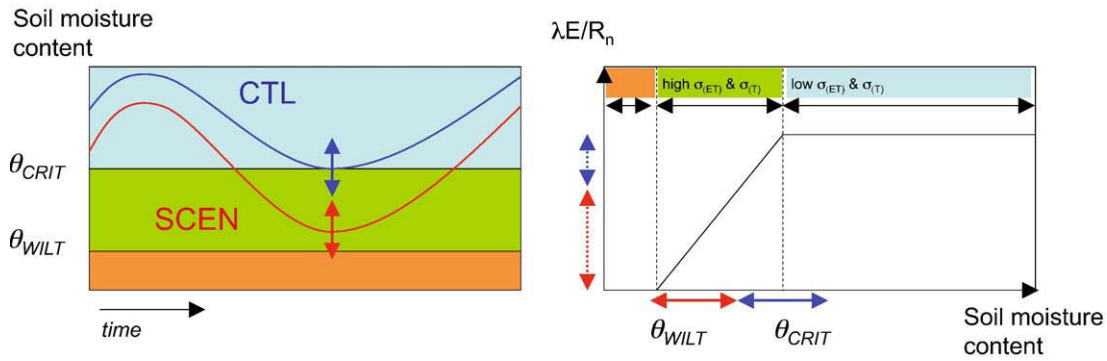


Fig. 13. Mechanism leading to an increase in temperature variability under future-climate conditions in regions shifting from wet to transitional climate regime. Left: Mean seasonal cycles of soil moisture for the two regimes (CTL vs SCEN, i.e. present vs future). Right: Evaporative fraction as a function of soil moisture content (see also Fig. 5) and illustration of impact of the same variability of soil moisture for the resulting variability of evaporative fraction in the case of the two different soil moisture regimes (CTL vs SCEN).

IPCC scenarios for Central Europe (for actual output from climate scenarios, see e.g. Vidale et al., 2007). In the same figure (right), the conceptual dependency of evaporative fraction on soil moisture is displayed, following the conceptual framework discussed in Section 4.1 (Fig. 5). In this hypothetical scenario, we consider that the mean seasonal cycle of soil moisture in the present-climate conditions is largely confined within a range where it does generally not limit evapotranspiration (radiation-limited evapotranspiration regime), and that it shifts to a soil moisture-limiting regime during the summer season under the future-climate conditions (blue vs red arrows in Fig. 13 right).

As can be inferred from this figure, the shift to a soil moisture-limiting regime means that evapotranspiration variability strongly increases under the climate-change scenarios. Consequently, the variability of sensible heat flux and air temperature is also expected to increase. This is consistent with the results from the analysed IPCC AR4 scenarios (Fig. 12), which show that the projected increase in summer temperature variability over the European continent is concomitant with a high increase of summer evapotranspiration variability over the same region. Indeed, regional climate simulations removing the interannual variability of soil moisture in climate-change scenarios display significantly (–60%) reduced summer temperature variability (Seneviratne et al., 2006a, Fig. 14).

One may wonder if changes in soil moisture regimes would also impact precipitation variability. Provided that a positive feedback between evapotranspiration and precipitation exists, increases in evapotranspiration variability may indeed result in enhanced precipitation variability. Because of the uncertainties associated with soil moisture–precipitation feedbacks (Section 5.2), it is difficult to infer whether such a mechanism may play an important role in reality. In a modelling framework, however, climate scenarios do present an enhancement of precipitation variability and extremes over Central and Eastern Europe despite the actual projected decrease of mean precipitation (e.g. Fig. 12; see also Christensen and Christensen, 2003; Frei et al., 2006; Seneviratne et al., 2006a). There is some indication that part of the projected enhancement of precipitation variability in this region may be associated with soil moisture feedbacks, though changes in atmospheric moisture content need to be taken into account.

7.3. Future hot spots of soil moisture–atmosphere coupling

As discussed in the preceding sections, climate-change projections suggest that major changes in soil moisture regimes will take place in some regions of the globe. This means that the location of transitional climate regions, and thus of hot spots of soil moisture–climate coupling, is not static but will be displaced by climate change.

Using the $\rho_{(E,T)}$ measure of soil moisture–temperature coupling (Appendix B.2), the location of future hot spots can be diagnosed from

IPCC AR4 projections (Seneviratne et al., 2006a). The resulting figure (Fig. 8f) exhibits some similarities with the diagnosed $\rho_{(E,T)}$ values for present climate (Fig. 8e). However, a strong shift can be identified over Central Europe, and to a lesser extent over the North-Eastern United States. This corresponds to a northward shift of climate regimes in the Northern Hemisphere, in effect consistent with the projected changes in radiation, temperature, and soil moisture. Further analyses should investigate to which extent similar features can be identified for the austral summer.

Shifts of climate and soil moisture regimes, and consequent impacts on the location of hot spots of soil moisture–atmosphere coupling, are not only relevant in the context of enhanced greenhouse gas concentrations, but indeed for any modification of the climate system resulting in significant soil moisture changes in given regions. These shifts may also be induced by other regional and/or global forcings, such as changes in aerosol concentrations (linked with dimming and brightening, e.g. Wild et al., 2005; Pinker et al., 2005; Philipona et al., 2009; see for instance Roderick and Farquhar, 2002; Robock and Li, 2006; Hobbins et al., 2008; Teuling et al., 2009; or with volcanic eruptions: e.g. Gu et al., 2003; Farquhar and Roderick, 2003), changes in large-scale circulation or SST patterns (e.g. Koster et al., 2009c), as well as land use changes (e.g. Haxeltine and Prentice, 1996; Ramankutty and Foley, 1999; Klein Goldewijk, 2001; Soares-Filho et al., 2006; Davin et al., 2007; Bonan, 2008; Pitman et al., 2009).

7.4. Are soil moisture–climate interactions amplifying or damping climate variability and anthropogenic climate change?

An important question regarding land–climate interactions is whether they lead to a damping or amplification of climate variability, particularly in the context of climate change. We have highlighted in the previous subsections how soil moisture–climate interactions increase climate variability in transitional climate zones, and thus increase climate variability in regions shifting to this climate regime. However, it is important to recall that under certain climate regimes, soil moisture may rather lead to a damping of the effects of precipitation or radiation variability on climate variables (soil moisture storage acting as a low-pass filter). Modified variability may result from reaching critical soil moisture thresholds, which play a role both at the low end (dry conditions) and at the high end (saturation).

Reconsidering Figs. 5 and 13, one can see that the (first-order) impact of soil moisture on climate variability depends as follows on the soil moisture regime:

(1) Wet soil moisture regime ($\theta \geq \theta_{CRIT}$):

Under this soil moisture regime, evapotranspiration is not sensitive to soil moisture, and soil moisture has no effect (or in some cases a

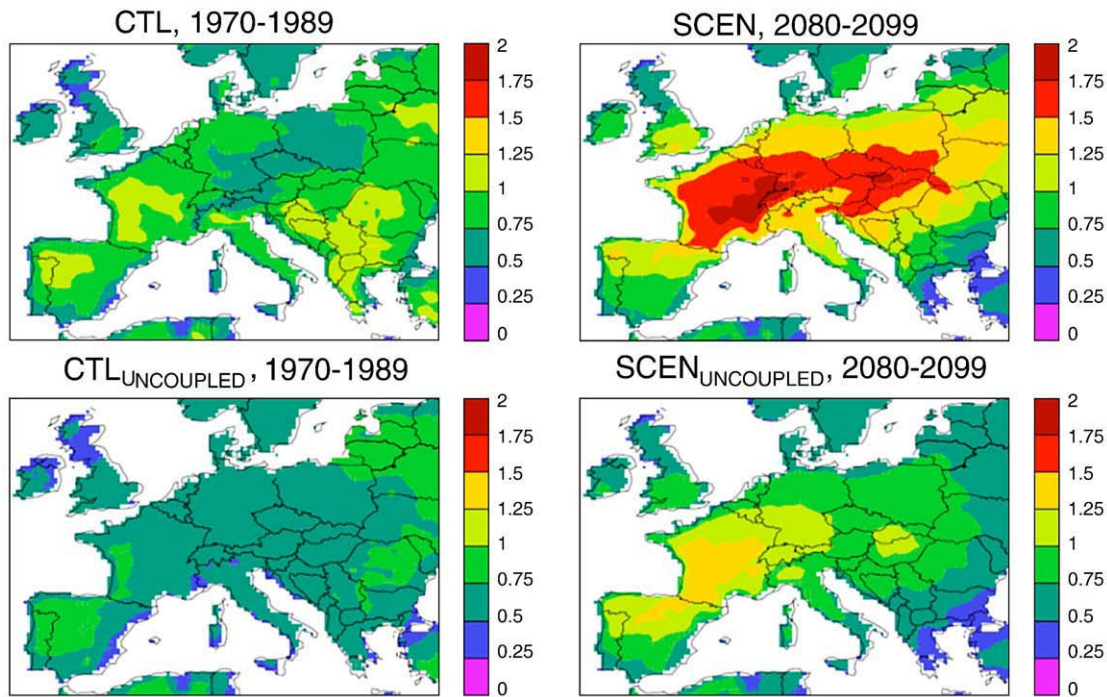


Fig. 14. Role of interannual variability of JJA soil moisture for interannual variability of JJA temperature (standard deviation) in Europe in present (left) and future (right) climate: (top) Reference (coupled) regional climate simulations (CHRM model) driven with PRUDENCE scenario (20C3M/SRES A2 and HadCM3–HadAM3H); (bottom) corresponding (uncoupled) regional climate simulations with prescribed soil moisture (climatology). The plots display the standard deviation of JJA temperature [K] in the four simulations. [Figure adapted from Seneviratne et al., 2006a].

damping effect⁷) on climate variability. However, runoff is sensitive to soil moisture content under this regime, as soil moisture is closer to its maximum threshold (saturation): High precipitation variability may lead to high streamflow variability that cannot be damped by soil moisture storage. When soil moisture is close to saturation, small variations in precipitation may thus lead to large variations in runoff.

(2) Transitional soil moisture regime ($\theta_{\text{WILT}} < \theta < \theta_{\text{CRIT}}$):

Under this soil moisture regime, evapotranspiration is sensitive to soil moisture and its variations are large enough to impact climate significantly (Section 4). Variations in precipitation and radiation forcing are directly reflected in variations of evapotranspiration and sensible heat flux, with strong variations of the Bowen ratio. Under this regime, variations in soil moisture are central in controlling climate variability, as discussed in the preceding subsections. This includes potential positive feedbacks with temperature and precipitation. Thus, under this regime soil moisture variations strongly contribute to climate variability and may even enhance it.

(3) Dry soil moisture regime ($\theta \leq \theta_{\text{WILT}}$):

Under this soil moisture regime, evapotranspiration is very sensitive to soil moisture but very limited due to the dry conditions (evapotranspiration is small and so are its variations; see Section 4, and Figs. 5, 6 and 13). Thus, the impact of variations in precipitation and radiation on climate variability is almost independent of the soil moisture content. Single large anomalies of precipitation may nonetheless have some impact on the climate given the large sensitivity of evapotranspiration on soil moisture. But heavy precipitation events on dry soils can also directly lead to runoff without infiltration of moisture due to modified soil properties.

⁷ For example, a positive radiation anomaly will lead to enhanced evaporation (and evaporative cooling) under this regime, effectively reducing the net resulting temperature anomaly and thus damping the warming effect of the radiation anomaly.

Regarding potential modifications under anthropogenic climate change, it is clear that shifts in soil moisture regimes may both lead to an *enhancement* or *decrease* of variability, for instance if the climate in one region shifts from a wet to a transitional soil moisture regime, or from a transitional to a dry soil moisture regime, respectively. As discussed in the preceding subsections, a good example for the first case is the projected increase in summer temperature variability in Central Europe. As a possible example for the second case, one can consider in Fig. 12 the Southeastern United States, where a mean decrease of soil moisture and concomitant decrease in soil moisture and temperature variability is projected, though it is difficult to separate causes and effects between the decrease in soil moisture variability (Fig. 12b) and the decrease in precipitation variability (Fig. 12f). Thus, decreases in soil moisture may both lead to an increase or decrease in climate (temperature) variability, depending on the original climate (and soil moisture) regime. Note that the links between soil moisture regime shifts and corresponding impacts on climate variability suggest some parallels with existing theory regarding critical transitions in complex dynamical systems (Scheffer et al., 2009).

8. Validation datasets for soil moisture

A major impediment for the investigation of soil moisture–climate interactions is the scarcity of soil moisture observations. Ground observations are cost-intensive and require important efforts to be put in place. As a consequence, only few ground-based soil moisture measurement networks are available (e.g. Robock et al., 2000). Remote sensing measurements, which provide indirect estimates of soil moisture (e.g. microwave remote sensing and GRACE satellite mission), represent some alternatives to intensive ground measurements, however they have several limitations.

Despite these issues significant progress has been accomplished in the field of soil moisture measurement (both ground-based and remotely-sensed) in recent years, as testify the growing number of publications on new measurement techniques and their comparison

(e.g. Walker et al., 2004; Tapley et al., 2004; Drusch et al., 2004; Reichle et al., 2004; Evett and Parkin, 2005; Blonquist et al., 2005; Andersen et al., 2005; Loescher et al., 2007; Wagner et al., 2007; Kerr and Le Vine, 2008; Gruhier et al., 2008; Robinson et al., 2008; Vereecken et al., 2008; de Jeu et al., 2008). Moreover, diagnostic methods, either using a combination of existing observations (e.g. Seneviratne et al., 2004; Hirschi et al., 2006a,b; Mueller et al., in press), offline land surface model simulations with observations-based forcing (e.g. Dirmeyer et al., 1999; Rodell et al., 2004a,b; Dirmeyer et al., 2006a; Peters-Lidard et al., 2007), or land data assimilation techniques (e.g. van den Hurk et al., 1997; Margulis et al., 2002; Crow and Wood, 2003; Reichle and Koster, 2005; Dunne and Entekhabi, 2006; Reichle, 2008), offer promising perspectives.

In this section, we will describe several methods to measure or estimate soil moisture, with their respective limitations. A summary is provided in Table 1. Section 8.6 and Table 2 also briefly present methods to measure or estimate evapotranspiration. Fig. 15 displays the geographical distribution of some of the mentioned datasets or of relevant variables for their accuracy.

8.1. Ground measurements

Several techniques are available to measure soil moisture in situ. We provide here a brief overview on the most common techniques. For more details and references, please refer to e.g. Hillel (1998), Robock et al. (2000), Walker et al. (2004), Evett and Parkin (2005), Blonquist et al. (2005), Robinson et al. (2008), and Vereecken et al. (2008). The distribution of stations from the Global Soil Moisture Data Bank (Robock et al., 2000) is displayed in Fig. 15a. This data bank includes data from most large-scale and long-term ground soil moisture measurement networks.

8.1.1. Gravimetric measurements

The only technique measuring soil water content per se is the gravimetric technique (see e.g. Robock et al., 2000, and Robinson et al., 2008 for an overview). Thereby, soil samples are extracted from the field, weighted before and after drying (usually in an oven at 105 °C). From the changes in mass, the original (absolute, relative, as well as volumetric) soil moisture content can be derived. While this method represents the actual “ground truth” (for single samples), there are several issues with its implementation. The most important is that the measurement method is destructive, and can thus not be reproduced. Moreover, significant manpower is required for the retrieval of the samples and the lab measurements. For these reasons, the temporal resolution of long-term measurement networks using this technique is usually coarse, typically of the order of 1–2 weeks at best. Nonetheless, gravimetric measurements still represent the reference measurements for calibrating other soil moisture measurement methods, since all other approaches are indirect. Several historical soil moisture datasets are based on gravimetric measurements (e.g. Robock et al., 2000), in particular in the former Soviet Union (e.g. Vinnikov and Yesserkepova, 1991).

8.1.2. Indirect in-situ measurement methods

There exist several indirect methods for in-situ measurements of soil moisture. Two of the most common techniques are either based on time domain reflectometry (TDR; see e.g. Topp and Reynolds, 1998; Robinson et al., 2003; Topp, 2003; Robinson et al., 2008) or soil capacitance measurements (e.g. Dean et al., 1987; Topp, 2003; Bogaena et al., 2007; Robinson et al., 2008). Both are electromagnetic techniques, which make use of the dependency of the dielectric permittivity of the soil on soil moisture content, i.e. the great difference between the dielectric constant of soil components and water, respectively. The TDR sensors, which operate at higher frequencies, are significantly more accurate than capacitance sensors (e.g. Robinson et al., 2008). However, the latter are of much lower cost, which can allow a higher number of instruments and thus much denser networks (e.g. Bogaena et al., 2007). Given the strong spatial heterogeneity of soil moisture properties, it can appear

attractive for some applications to choose slightly less accurate but cheaper sensors in order to increase the spatial sampling (see e.g. Bogaena et al., 2007; Robinson et al., 2008). But their low accuracy for wet conditions, remains a significant issue for the interpretation of the data.

Other indirect sensors that have been used in some soil moisture measurement networks include neutron probes (e.g. Hollinger and Isard, 1994; see also Hillel, 1998; Robinson et al., 2008). Though these sensors are fairly accurate, they are not used as often as for instance TDR and capacitance sensors, due to the health risks associated with the radioactive elements used as source for the neutrons (Hillel, 1998; Robinson et al., 2008).

Further indirect soil moisture measurement techniques include electrical resistance measurements, heat pulse sensors, fiber optic sensors, and gamma ray scanners. For more details on these methods and references to relevant publications see e.g. Hillel (1998), Robock et al. (2000), Robinson et al. (2008), and Vereecken et al. (2008). Some networks also use tensiometers, but the conversion of the measured soil suction to volumetric soil moisture content (e.g. Brooks and Corey, 1964; van Genuchten, 1980; Kosugi, 1994) strongly depends on soil constituents and is thus associated with fairly large uncertainties (e.g. Hillel, 1998), in particular for dry conditions.

8.2. Remote sensing measurements

In recent years, major advances have taken place in the field of soil moisture remote sensing. Remote sensing methods include passive and active microwave remote sensing (radiometers, scatterometers, and synthetic aperture radars), and gravity recovery measurements (for an overview of techniques and relevant publications, see e.g. Schmugge et al., 2002; Tapley et al., 2004; Wagner et al., 2007; Kerr, 2007; Robinson et al., 2008; Vereecken et al., 2008; de Jeu et al., 2008). Some other techniques have also been recently investigated, such as the use of GPS receivers (Larson et al., 2008). Here we will present a brief overview on microwave remote sensing, and the Gravity Recovery and Climate Experiment (GRACE) satellite mission.

8.2.1. Microwave remote sensing

Excellent reviews on microwave remote sensing techniques are provided in e.g. Schmugge et al. (2002), Wigneron et al. (2003), Jackson (2005), Wagner et al. (2007), and de Jeu et al. (2008). As is the case for TDR and capacitance sensors, microwave remote sensing techniques make use of the fact that the soil dielectric constant increases with increasing water content (e.g. Ulaby et al., 1982; Wagner et al., 2007). Both active and passive microwave remote sensing are commonly used to retrieve soil moisture information. Passive microwave methods measure the natural emission of the land surface, while active microwave sensors measure the energy scattered back from the land surface after transmission of an electromagnetic pulse (e.g. Schmugge et al., 2002; Wagner et al., 2007).

Several satellite measurement missions have been launched for microwave remote sensing from space. Some of the major missions in radiometry (passive sensors) include the Scanning Multichannel Microwave Radiometer (SMMR, 1978–1987; resolution between 27 km and 148 km depending on frequency), the Advanced Microwave Scanning Radiometer (AMSR-E, launched in 2002; resolution between 38 km and 56 km), and the new Soil Moisture and Oceanic Salinity (SMOS; e.g. Kerr et al., 2001; Kerr, 2007) mission from the European Space Agency (ESA), which was launched in November 2009 (ground resolution between 30 km and 50 km). While SMMR and AMSR-E operated in the C-Band (frequency $f=4-8$ GHz, wavelength $\lambda=7.5-3.8$ cm), SMOS operates in the L-Band ($f=1-2$ GHz, $\lambda=30-15$ cm), which is suggested to be more appropriate for soil moisture applications (e.g. Wagner et al., 2007). Global soil moisture datasets based on SMMR and AMSR-E have been recently derived (e.g. Owe et al., 2001; de Jeu, 2003; Njoku et al., 2003; Owe et al., 2008; de Jeu et al., 2008). Additional soil moisture datasets were also derived from sensors with higher

Table 1

Overview on available soil moisture (and terrestrial water storage) estimates.

	Ground measurements	Atmospheric–terrestrial water-balance estimates (BSWB)	GRACE	Microwave remote sensing (radiometers and scatterometers)	LSM with observation-based forcing
Spatial resolution	Point measurements	300–1000 km (10^5 – 10^6 km ²)	500–1000 km	30–50 km	Typically 10–50 km, some applications down to 1km for selected regions
Measurement depth	Variable; whole profile can be measured	Whole terrestrial water storage	Whole terrestrial water storage	Top centimeters of soil	Model estimate
Geographical coverage	Point measurements; in some cases regional networks (long-term datasets: Illinois, former Soviet Union, Mongolia, China)	Only derivable in regions with runoff observations	Global coverage	Global coverage but not suitable in regions with dense vegetation cover	Global coverage, but regional quality depends on the availability of corresponding measurements of relevant forcing variables
Temporal resolution	Manual retrievals: biweekly or monthly measurements; automatic retrievals: resolve diurnal cycle	Monthly estimates; submonthly resolution may be possible	Monthly estimates	Every 1–4 days	Resolves diurnal cycle
Temporal coverage	Very few long-term datasets (>5 years); end of measurements after 1990 in many areas of the former Soviet Union; networks with automatic retrievals only for a few years	Retrospective estimates can be derived over the whole length of reanalysis datasets (ca. 1950–present, depending on the product); the availability of recent estimates is constrained by the availability of runoff data	2002–present	SMMR (1978–1987); ERS-1 and ERS-2 (1991–); TMI (1997–); AMSR-E (2002–); WindSat (2003–); ASCAT (2006–); SMOS (2009–); SMAP (launch planned in 2015)	Depends on availability of forcing, typically reanalysis data period (1950–present)
Main advantage	Reference measurements; gravimetric measurements; direct measurement of soil moisture per se; TDR and capacitance sensors; automatic retrievals allow high temporal resolution; when used in profiles, can provide detailed depth-dependent information	Retrospective dataset back to ca. 1950; large coverage	Global coverage	Coverage in regions without ground observations	Global coverage is possible; relatively high resolution; retrospective datasets back to ca. 1950
Main disadvantage	Few observations; point-scale measurements; expensive and requires maintenance; gravimetric measurements: destructive and cannot be reproduced at same location; capacitance sensors: low accuracy for wet conditions	Dependent on quality of convergence data (changes in assimilated data; drifts); cannot be used to detect trends; low resolution	Only recent data; low resolution	Discontinuous datasets; only top soil moisture; no data under dense vegetation cover	Results dependent on quality of forcing data; models tuned for regions with observations; discrepancies between models (water-holding capacity, soil moisture definition)

Table 2
Overview on available evapotranspiration estimates.

	Ground measurements: lysimeters	Ground measurements: eddy-covariance measurements	Terrestrial water balance	Atmospheric water-balance estimates (BSWB)	LSM with observation-based forcing
Spatial resolution	Point measurements	Point measurements	Catchment scale	300–1000 km (10^5 – 10^6 km ²)	Typically 10–50 km, some applications down to 1 km for selected regions
Geographical coverage	Point measurements; very few measurement locations	Point measurements (footprint): FLUXNET network well represented in North America, Europe, Australia, Eastern Asia	Depends on availability of runoff and precipitation measurements	Can be derived in all regions with good precipitation measurements	Global coverage, but regional quality depends on the availability of corresponding measurements of relevant forcing variables
Temporal resolution	Resolves diurnal cycle	Resolves diurnal cycle	Yearly sums	Monthly estimates; submonthly resolution may be possible	Resolves diurnal cycle
Temporal coverage	Very few long-term datasets (started in ~1970s)	Most measurements started after 2000	Depends on availability of runoff and precipitation measurements	Retrospective estimates can be derived over the whole length of reanalysis datasets (ca. 1950–present, depending on the product); the availability of estimates is only constrained by the availability of precipitation data	Depends on availability of forcing, typically reanalysis data period (1950–present)
Main advantage	Most accurate measurement method	Ground reference measurements; often concomitant with comprehensive ecosystem measurements	Good availability of data in many regions; can be used to derive long-term trends	Retrospective dataset back to ca. 1950; large coverage	Global coverage is possible; relatively high resolution; retrospective datasets back to ca. 1950
Main disadvantage	Few observations; point-scale measurements; expensive and requires maintenance	Issues with energy balance closure; not applicable in inhomogeneous terrain; variable footprint; expensive and requires maintenance	Assumes no annual changes in storage; yearly sums only	Dependent on quality of convergence data (changes in assimilated data; drifts); cannot be used to detect trends; low resolution	Results dependent on quality of forcing data; models tuned for regions with observations; discrepancies between models (sensitivity on soil moisture, parameterizations)

frequency ($f > 10$ GHz), for which however, vegetation effects are larger (Wagner et al., 2007). Some examples include datasets based on the Tropical Rainfall Measuring Mission (TRMM) Microwave Imager (TMI) X-Band (10.65 GHz) radiometer (measurements since 1997; Gao et al., 2006; Drusch, 2007), and from the WindSat radiometer (measurements since 2003; using the 10.7, 18.7 and 37 GHz V and H channels, Li et al., in press). Further applications based on retrievals from the Special Sensor Microwave Imager (SSM/I, lowest frequency at 19.35 GHz, 1987–present) are also discussed in e.g. Wagner et al. (2007).

Besides radiometers, other space-borne sensors that can be used for microwave remote sensing of soil moisture are scatterometers (active sensors), provided that they operate in the appropriate frequency. Datasets have been for instance derived from scatterometers on board of the European Remote Sensing satellites ERS-1 (1991–2000) and ERS-2 (since 1995), and from the new Advanced Scatterometer (ASCAT, since 2006) on board the Meteorological Operational (MetOp) satellites (e.g. Scipal, 2002; Wagner et al., 2003; Bartalis et al., 2007; de Jeu et al., 2008). An additional mission, the Soil Moisture Active and Passive (SMAP) mission, which will include both active and passive sensors (in the L-Band), is currently planned for launch in 2015 (<http://smap.jpl.nasa.gov/>).

The main limitations associated with microwave remote sensing are that only surface soil moisture can be retrieved (a few centimeters at most, e.g. Kerr, 2007; Wagner et al., 2007), while it is root-zone soil moisture that is relevant for most climate applications (Sections 3 and 4). Under dry conditions, surface soil moisture might become “decoupled” from the root-zone soil moisture, limiting any approach to derive root-zone soil moisture from surface observations (Capehart and Carlson, 1997). Moreover, measurements are difficult under dense vegetation cover, which excludes several regions of the world with potential important soil moisture effects (e.g. Drusch et al., 2001; Reichle et al., 2004; Wagner et al., 2007; de Jeu et al., 2008). Finally, it is important to note the large uncertainties remaining in the development of corresponding retrieval algorithms, leading to significant differences between products for different (or even the same) satellite missions (e.g. Reichle et al., 2007; Rüdiger et al., 2009). Nonetheless, recent applications have illustrated the potential of microwave remote sensing data for data assimilation and monitoring (Reichle et al., 2007; Loew et al., 2009; Mueller et al., in press).

Fig. 15c displays the global distribution of vegetation densities (or vegetation optical depth τ , which is directly proportional to the vegetation water content), based on an estimate of Meesters et al. (2005) (redrawn from de Jeu et al., 2008, Fig. 5b). This latter estimate is useful to assess the applicability of microwave remote sensing techniques, which are less accurate for dense vegetation cover. As presented in de Jeu et al. (2008), the correlation between two independent datasets based on this technique (using AMSR-E and ERS estimates), has shown the best agreement in regions with $\tau = 0.1$ – 0.5 . Correlations strongly decrease for $\tau > 0.5$, and in particular $\tau > 0.6$, suggesting less reliability of the technique for such vegetation densities (possibly also due to issues with current retrieval algorithms). For deserts ($\tau < 0.1$), the two independent datasets do not present any strong correlations, although they both exhibit low values. It is expected that the new SMOS satellite sensor will provide more reliable estimates for vegetation densities corresponding to $\tau > 0.5$, because it operates in L-band, which has a deeper vegetation penetration.

8.2.2. Gravity Recovery and Climate Experiment (GRACE)

The twin satellites of the Gravity Recovery and Climate Experiment (GRACE), launched in 2002, allow the derivation of the Earth's gravity field with high accuracy and with approximately monthly temporal resolution (e.g. Tapley et al., 2004; Wahr et al., 2004; Ramillien et al., 2008). Seasonal variations in terrestrial water storage (soil moisture, groundwater, snow, surface water, and ice cover) result in variations of the Earth's gravity field, which can be thus measured with GRACE.

Several studies have shown that large-scale (ca. 500–1000 km) estimates of variations in terrestrial water storage can effectively be

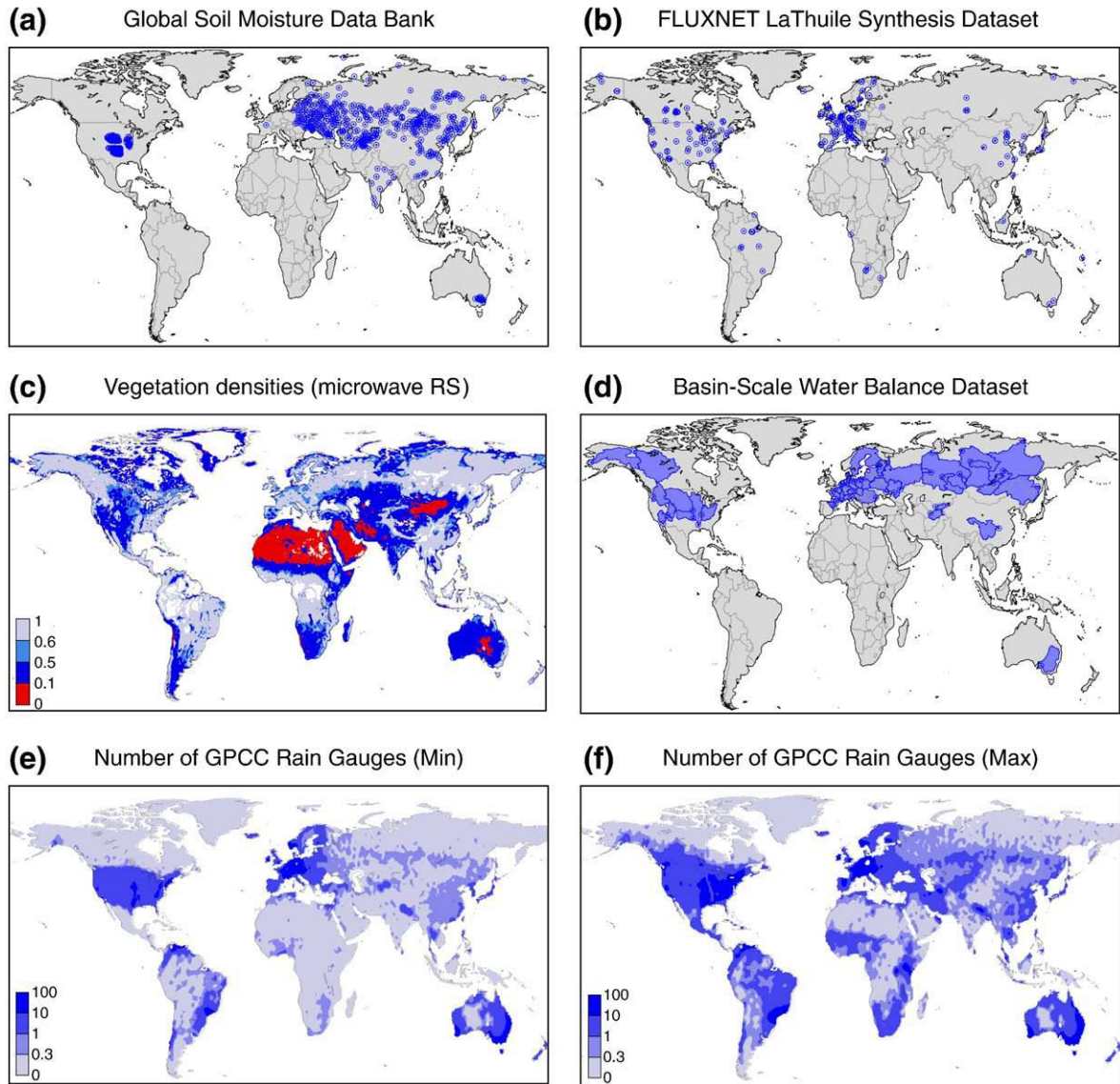


Fig. 15. Spatial coverage of measurements or of variables impacting the reliability of measurements/estimates: (a) available long-term ground observations of soil moisture from the Global Soil Moisture Data Bank, (b) FLUXNET measurement sites from LaThuile Synthesis Dataset (2007), (c) vegetation densities (relevant for microwave remote sensing), (d) BSWB estimates of monthly changes in terrestrial water storage, (e,f) Minimum (e) and maximum (f) density of GPCC rain gauges in the period 1986–1995 (monthly product from ISLSCP2, number gauges per $1^\circ \times 1^\circ$ grid cell in the period). A weak spatial smoothing has been applied in (e,f) for display purposes. [Sources: (a) from Robock et al., 2000; (b) FLUXNET LaThuile Synthesis Dataset; (c) Redrawn from de Jeu et al., 2008; (d) ETH Zurich, BSWB dataset; (e, f) data derived from 1986 to 1995 GPCC dataset at ISLSCP2 archive (http://islscp2.sesda.com/ISLSCP2_1/html_pages/groups/hyd/gpcc_precip_monthly_xdeg.html)].

retrieved from GRACE measurements (e.g. Tapley et al., 2004; Wahr et al., 2004; Rodell et al., 2004b; Andersen et al., 2005; Schmidt et al., 2006; Swenson et al., 2006; Güntner et al. 2007; Swenson et al. 2008; Rodell et al., 2009). Two main issues with GRACE estimates are, however: 1) the necessary disaggregation of the data in separate estimates of the individual terrestrial water storage components (e.g. soil moisture, groundwater, snow); 2) the still coarse resolution of the estimates (ca. 400–500 km at best, see e.g. Rowlands et al., 2005; Swenson and Wahr, 2006; Chen et al., 2006).

8.3. Atmospheric–terrestrial water-balance estimates and BSWB dataset

Following the combined atmospheric–terrestrial water-balance approach (e.g., Rasmusson, 1968; Oki et al., 1995; Masuda et al., 2001; Seneviratne et al., 2004; Hirschi et al., 2006a,b; Mueller et al., in press), the monthly terrestrial water storage variations in a given river basin can be derived from the combined atmospheric and terrestrial water-balance equations. The terrestrial water-balance equation is provided

in Section 3 (Eq. (5)), and the atmospheric water-balance equation is expressed as follows:

$$\left\{ \frac{\partial W}{\partial t} \right\} = - \left\{ \nabla_H \cdot \vec{Q} \right\} - \{ \bar{P} \} + \{ \bar{E} \} \quad (12)$$

where W represents the column storage of atmospheric water vapour, $\vec{Q}$ the vertically integrated two-dimensional atmospheric water-vapour flux, P the precipitation, and E the evapotranspiration. The operator $(\nabla_H \cdot)$ represents the horizontal divergence, the overbar a temporal average (i.e., monthly means) and $\{ \}$ a space average over the region.

Eqs. (5) and (12) can be then combined to obtain the following expression:

$$\left\{ \frac{\partial S}{\partial t} \right\} = - \left\{ \frac{\partial W}{\partial t} \right\} - \left\{ \nabla_H \cdot \vec{Q} \right\} - \{ \bar{R} \} \quad (13)$$

where S represents the total terrestrial water storage (soil moisture, groundwater, snow, surface water, and ice cover) of the area, and R is the measured streamflow (assumed to include both the surface and the groundwater runoff of the area). Recent applications have shown promising results for estimating monthly changes in terrestrial water storage with Eq. (13), using observational runoff data, and reanalysis or operational forecast analysis data for the terms $\left\{\frac{\partial W}{\partial t}\right\}$ and $\left\{\nabla_H \cdot \vec{Q}\right\}$ (Seneviratne et al., 2004; Hirschi et al., 2006a,b; Troch et al., 2007; Mueller et al., in press). One main advantage of using $\left\{\frac{\partial W}{\partial t}\right\}$ and $\left\{\nabla_H \cdot \vec{Q}\right\}$ from reanalysis datasets is that these terms depend to a large extent on assimilated variables (radiosonde measurements of moisture and wind). Thus it can be shown that the estimates of terrestrial water storage derived with Eq. (13) have a much better agreement with observations than soil moisture output from the reanalysis datasets, which are mostly a model product (Seneviratne et al., 2004). Indeed they only depend on assimilated variables and streamflow measurements. Uncertainty estimates of the respective terms of Eq. (13) are provided in Mueller et al. (in press).

The spatial scale at which Eq. (13) can be applied is of the order of 300–1000 km (10^5 km²– 10^6 km²). The raw terrestrial water storage variations estimates derived with Eq. (13) are generally characterized by a long-term drift (Seneviratne et al., 2004; van den Hurk et al., 2005; Hirschi et al., 2006a), likely due to small systematic errors in the atmospheric moisture flux convergence from the employed reanalysis (see also above mentioned publications). This drift can be removed using various detrending approaches (e.g., removal of running mean over a given time window).

The Basin-Scale Water Balance (BSWB) dataset from ETH Zurich (http://iacweb.ethz.ch/data/water_balance/) provides estimates of monthly changes in terrestrial water storage derived with Eq. (13) using common reanalysis or operational forecast analysis data products (ERA40, ECMWF operational analysis, North American Regional Reanalysis, ERA-Interim) and streamflow observations (Seneviratne et al., 2004; Hirschi et al., 2006a,b; Mueller et al., in press). Fig. 15d displays the river basins for which BSWB estimates have been derived. The BSWB estimates have been shown to perform well compared to other terrestrial datasets, e.g. ground observations, as well as GRACE- and ERS-scatterometers based estimates. Their main advantage is that using existing runoff observations and reanalysis datasets, retrospective estimates can be derived for several decades up to present. The main limitations of the approach are related to its coarse scale, the fact that it provides (similar to GRACE) only estimates of total terrestrial water storage and not of its single components (soil moisture, groundwater, and snow), and that the required detrending impairs the use of the derived estimates to investigate long-term trends in the hydrological cycle. However, the derived BSWB dataset can provide valuable information on the mean seasonal cycle and interannual variations in terrestrial water storage of worldwide river basins over several decades.

8.4. LSM-based estimates

As seen above, there are at present no comprehensive and global datasets of observed soil moisture, since ground observations are very limited in space, remote sensing estimates have limitations regarding temporal and/or spatial coverage, accuracy and their exact link with root-zone soil moisture, and atmospheric–terrestrial water-balance estimates are available only at coarse scale and in river basins with observed runoff (with the mentioned issues regarding long-term trends). Thus, output from land surface models, driven with observation-based data can be a useful alternative to derive global

soil moisture datasets. Two such products are the Global Land Data Assimilation System dataset (GLDAS, Rodell et al., 2004a), available from 1979 to present (but driven with heterogeneous forcing datasets over time), and the Global Soil Wetness Project 2 dataset (GSWP-2, Dirmeyer et al., 2006a), available for the time period 1986–1995 (a previous dataset for 1987–1988 is also available, Dirmeyer et al., 1999). Efforts are also being invested to develop an offline LSM product with high spatial (1 km) and temporal resolution (the Land Information System or LIS, Kumar et al., 2006; Peters-Lidard et al., 2007). Moreover, also new forcing datasets that cover longer time periods are being developed (e.g. Sheffield and Wood, 2008).

Analyses have shown that the GSWP-2 dataset compares reasonably well with available observations, with the multi-model average generally outperforming any single model (Guo et al., 2007). Reasons for this result may lie in compensation of the individual model biases. Due to the lack of ground validation datasets, there are, however, several regions where the quality of such datasets cannot be verified.

One issue that is often put forth in LSM-computed datasets is the dependency of the obtained results on the quality of the forcing datasets (precipitation, radiation; see e.g. Oki et al., 1999; Guo et al., 2006b). Oki et al. (1999) showed for instance that a density of at least 30 rain gauges per 10^6 km² is necessary for accurate runoff prediction. Assuming that this also applies to soil moisture estimates, Koster et al. (2004b) and Reichle et al. (2004) showed for the GPCP (Huffman et al., 1997) dataset, that this implies that appropriate estimates of soil moisture can in theory only be retrieved in North America, Europe, and part of China. We provide a similar analysis in Fig. 15e,f based on GPCC measurements for the time period 1986–1995. These plots display respectively the minimum and maximum number of GPCC rain gauges per $1^\circ \times 1^\circ$ grid box in the monthly ISLSCP2 product for 1986–1995. According to the estimate of Oki et al. (1999), the critical threshold for accurate land surface simulations would correspond to ca. 0.3 rain gauges per $1^\circ \times 1^\circ$ grid box. If one considers the minimum number of rain gauges (which constrains the reliability of long-term time series), it is clear that reliable data lack in most parts of the world (Fig. 15e), except in North America, Europe, part of Australia, and China. However, for shorter time spans, the coverage can be better (Fig. 15f). Fekete et al. (2004) similarly identified that there are strong divergences between existing precipitation datasets in regions with low density of precipitation measurements. Even though radiation is an additional important forcing variable for soil moisture trends, observed variations in e.g. incoming solar radiation (Wild et al., 2005) are often not taken into account in LSM datasets, and represent an added source of uncertainty.

8.5. Drought indices

Drought indices are computed based on meteorological data, i.e., precipitation observations, often in combination with temperature data. While they may rely on simple water-balance models (e.g. PDSI, see below), their purpose is to provide estimates of water scarcity rather than soil moisture per se. The main advantage of this approach is the high availability and good geographical and temporal coverage of meteorological observations.

A comprehensive review of drought indices can be found in Heim (2002). In the context of land–climate interactions, the most prominent indices are the Standard Precipitation Index (SPI) and the Palmer Drought Severity Index (PDSI). The SPI (McKee et al., 1993) is designed to quantify deficits in precipitation on different time scales, e.g., 3, 9 or 12 months. For a given location, the SPI is established from precipitation observations for a reference period. The observations are fitted to a probability distribution and transformed into a normal distribution. The mean SPI value for the reference period is thus zero. Positive (negative) values signify more (less) than average precipitation. The SPI calculated for time scales between 9 and 12 months has been found to be quasi equivalent to the PDSI (Lloyd-Hughes and Saunders, 2002). The PDSI

(Palmer, 1965) reflects local anomalies in the soil moisture supply. It is based on the analysis of precipitation and temperature data in a soil water balance. The resulting estimate of the temporal evolution of the soil moisture content is standardized to form an index that is positive (negative) for anomalously wet (dry) conditions. Some examples of applications using the PDSI index are for instance provided in the studies by Briffa et al. (1994) and Dai et al. (2004a,b). Further drought indices can also be derived using the PDSI model as basis (e.g. Corti et al., 2009).

8.6. Methods to measure and estimate evapotranspiration

In the investigation of soil moisture–climate interactions, evapotranspiration is a key variable for three reasons: 1) In the terrestrial water balance (Eq. (5)), soil moisture and evapotranspiration are the two main unknowns, and thus measurements of one of these two quantities can allow to derive estimates in the other (with rainfall and runoff measurements); 2) Concomitant evapotranspiration and soil moisture measurements are crucial to derive soil moisture–evapotranspiration relationships (Section 4); 3) In some instances, inferences regarding soil moisture control on evapotranspiration can be derived from evapotranspiration measurements alone (e.g. during dry-downs, Appendix B.3). We provide an overview on evapotranspiration measurement methods and approaches below (for a summary, see Table 2).

A standard approach to measure actual evapotranspiration on the ground is to use weighing lysimeters. Due to their costs, there are, however, only few instruments worldwide with long-term measurement records. Some exceptions are for instance the sites of Rietholzbach (<http://www.iac.ethz.ch/url/rietholzbach/>, in operation since 1976; Seneviratne et al., submitted for publication), and Rheindahlen (<http://www.niederrheinwasser.de>, since 1982).

Another well investigated and now common approach is to estimate evapotranspiration using eddy-covariance flux measurements, as is done as part of the FLUXNET network (e.g. Baldocchi et al., 2001). The present FLUXNET network includes more than 500 sites worldwide (<http://www.fluxnet.ornl.gov/fluxnet/index.cfm>), with excellent coverage in Europe and North America (e.g. Baldocchi, 2008; see Fig. 15b). However, one should note some issues with these measurements: 1) The turbulent fluxes are often underestimated (e.g. Foken, 2008); 2) The measurements are point-scale (small footprint area) and their interpretation for larger-scale areas is non-trivial; 3) The footprint area of which the measurements are representative depends on weather conditions and can thus vary in time; 4) Due to limitations of the approach, the measurements are generally performed in rather homogeneous flat terrain, and thus are not representative of all types of surface conditions (e.g. Rotach et al., 2004). Moreover, root-zone soil moisture observations are often lacking at FLUXNET sites (e.g. Granier et al., 2007). Nonetheless, the new FLUXNET synthesis dataset offers for the first time valuable observations of land hydrological variables, and should thus provide significant help in understanding soil moisture–climate relationships (e.g. Rambal et al., 2003; Ciais et al., 2005; Bréda et al., 2006; Granier et al., 2007; Reichstein et al., 2007; Baldocchi, 2008; Stöckli et al., 2008; Teuling et al., 2009; Jaeger et al., 2009).

Another approach to estimate evapotranspiration at the basin scale is to use atmospheric water-balance estimates by applying Eq. (12) (Section 8.3) and solving it for E:

$$\{E\} = \{P\} + \left\{ \overline{\nabla_H \cdot \vec{Q}} \right\} + \left\{ \frac{\partial W}{\partial t} \right\} \quad (14)$$

The estimates are then derived using e.g. reanalysis data for the terms $\left\{ \frac{\partial W}{\partial t} \right\}$ and $\left\{ \overline{\nabla_H \cdot \vec{Q}} \right\}$, and precipitation observations (e.g. Yeh et al., 1998; Hirschi et al., 2007; Jaeger et al., 2008; Yeh and Famiglietti, 2008). Note that similarly as for the estimates of terrestrial

water storage variations derived with Eq. (13), estimates derived with Eq. (14) are only marginally impacted by errors in the land surface model of the respective reanalysis dataset, since the employed atmospheric quantities are mostly based on assimilated data (radiosonde measurements). However, these estimates can again not be used for the investigation of long-term trends, due to the known artificial trends induced by the data assimilation systems. Further advantages and limitations of this approach are similar to the points discussed in Section 8.3.

When only evapotranspiration is of interest, also estimates using the terrestrial water balance and assuming zero change in soil moisture storage in annual mean can be derived using precipitation and runoff observations (e.g. Hobbins et al., 2004; Teuling et al., 2009). While only yearly estimates can be derived and the assumption of negligible yearly changes in storage cannot be assumed to hold in all regions (for instance due to human water use, e.g. Rodell et al., 2009, but also linked with soil moisture variations on longer time scales), this approach is nonetheless useful for some applications, in particular to investigate long-term trends. However, one should keep in mind that there are also non-negligible uncertainties in precipitation datasets (see discussion in Section 8.4 and Fig. 15e,f), and that runoff observations are lacking in several regions (Legates et al., 2005; Peel and McMahon, 2006).

Also land surface models can be used for the estimation of evapotranspiration for instance as part of the GLDAS (Rodell et al., 2004a) and GSWP-2 (Dirmeyer et al., 2006a) products. Though evapotranspiration from these products has been less investigated than soil moisture up to now, one can assume that similar issues apply for the estimation of evapotranspiration as for that of soil moisture, in particular the dependence on the quality of the atmospheric forcing (Section 8.4).

Modelling algorithms can also be applied for the derivation of evapotranspiration estimates from remote sensing data (e.g. Kustas and Norman, 1996; Anderson et al., 1997; Diak et al., 2004; McCabe and Wood, 2006; see also Kalma et al., 2008 and Jimenez et al., 2009 for overviews and assessments of existing datasets).

8.7. Scale issues of observational datasets

One particular issue of most datasets is their spatial resolution and coverage. Ground observations (e.g. TDR) are generally point measurements, confined within networks of limited extent, with necessarily scattered coverage. Satellite remote sensing observations (e.g. microwave remote sensing), on the other hand, are generally global, but of low resolution and often of poor quality on part of the globe depending on the measurement characteristics and limitations of the retrieval algorithms.

In hydrology, the spatial characteristics of observational datasets are often summarized with the “scale triplet” proposed by Blöschl and Sivapalan (1995), encompassing support, spacing and extent. The support refers to the “integration volume” of the measurement, the spacing to the distance between single measurements, and the extent to the area over which measurements are available (e.g. measurement network). However, a fourth “dimension” needs to be taken into account, i.e. how these characteristics vary globally (i.e. global distribution of corresponding measurement networks), possibly defining a “scale quadruplet”. For instance, there exists a limited number of soil moisture measurement networks over the globe (e.g. Vinnikov and Yezzerkepova, 1991; Hollinger and Isard, 1994; Robock et al., 2000), each using different sensors and/or measurement techniques, and with variable spacing and spatial extent. To illustrate current issues with the global coverage of validation datasets, we display in Fig. 15 maps related to the spatial coverage of different validation datasets mentioned in this section (see discussions in previous subsections).

Similarly to the spatial scale “quadruplet” discussed above, one should consider the following four characteristics to fully characterize the temporal coverage of validation datasets: 1) the amount of time over which the measurement is performed (equivalent to support in the spatial sense); 2) the temporal resolution of the measurements

(equivalent to spacing); 3) the time period that the measurements of a given network cover; 4) the overall temporal coverage of all measurement networks based on similar techniques. Note that the “measurement network” can also be considered as corresponding to the ensemble of retrievals from a single satellite mission.

As we have seen in this section, the available validation datasets for soil moisture and/or evapotranspiration each cover distinct spatial and temporal scales as defined by the spatial and temporal scale “quadruplets” mentioned above. To illustrate this point, we present a schematic in Fig. 16, which summarizes the spatial and temporal characteristics of existing validation datasets for specific measurement/estimation approaches, by displaying the spatial and temporal scales that they can respectively resolve or cover (in the case of point-scale observations, the scales correspond to the most extended network using such observations, e.g. AmeriFlux for the eddy-covariance flux measurements). The shading of the different areas corresponds to the spatial continuity of coverage (scattered point observations vs. globally valid observations).

The mentioned scale discrepancy between available observational/estimation methods is often an issue for the comparison and systematic analysis of existing validation datasets. Several studies have investigated approaches to characterize the spatial variability and mean regional behaviour of point-scale soil moisture observations (e.g. Vachaud et al., 1985; Vinnikov et al., 1996; Robock et al., 1998; Grayson and Western, 1998; Entin et al., 2000; Western et al., 2002; Wilson et al., 2004; Robinson et al., 2008). In order to be able to characterize soil moisture–climate relationships at the regional scale, dense ground observational networks would be ideally needed, as variability in point-scale observations is large. While soil moisture fields exhibit a certain rank stability (e.g. Vachaud et al., 1985; Kachanoski and de Jong, 1988; Grayson and Western, 1998; Robinson et al., 2008), establishing relations between point-scale and regional soil moisture is problematic and these relations are also impacted by climate variability (Teuling et al., 2007). Similarly, the strength of coupling between evapotranspiration and soil moisture may strongly vary at the local scale (Famiglietti and Wood, 1994). Despite these issues, spatial scales of soil moisture variability may overall be divided in two categories (Vinnikov et al., 1996; Robock et al., 1998; Entin et al., 2000): the small-scale variability (less than 20 km), which is impacted by small-scale variations in surface and subsurface characteristics (soil characteristics, soil inhomogeneities, small-scale variations in land cover), and the regional-scale variability (50–400 km), impacted by meteorological/climatological forcing (weather systems, precipitation, and radiation). This point is illustrated in Fig. 17 (from Robock et al., 1998).

Thus, even with a moderately dense ground-measurement network, it should be possible to at least capture the relationships controlling the regional-scale variability of soil moisture. Similarly, analyses from point-scale observations in Illinois have shown that, though absolute soil moisture content can be highly variable between neighbouring sites, the dynamic evolution of soil moisture (seasonal soil moisture changes) are often closely related (see e.g. Seneviratne, 2003). It is thus the dynamics of soil moisture at the regional scale that should be compared between validation datasets based on different sensors or estimation approaches.

Given the heterogeneity of available soil moisture (and evapotranspiration) datasets and the fact that no ideal dataset exists, the best approach may be to merge existing information, e.g. with land data assimilation. While first results in this field suggest some potential (e.g. van den Hurk et al., 1997; Margulis et al., 2002; Crow and Wood, 2003; Reichle and Koster, 2005; Ni-Meister et al., 2006; Dunne and Entekhabi, 2006; Muñoz Sabater et al., 2007; Balsamo et al., 2007; Zaitchik et al., 2008; see also Reichle, 2008 for a recent review), large uncertainties remain. One issue is again the intrinsic limitations of the estimates of soil moisture that are assimilated (e.g. top soil moisture in the case of microwave remote sensing, which then

needs to be mapped to root-zone soil moisture, e.g. Calvet and Noilhan, 2000; Muñoz Sabater et al., 2007). Another key issue is the scarcity of ground-based observations. Any estimation, indirect measurement, or data assimilation approach, needs indeed to be validated with such ground datasets.

Finally, one should also note that the issue of scales is intrinsically related to the question that needs to be addressed with the analysis of the respective datasets. For instance, for investigating impacts of soil moisture for convective events, the resolution of the data (both in space and time) is more important than its continuity or length. For investigations related to subseasonal to seasonal forecasting, multi-year datasets for larger regions are needed. Finally, for climate-related issues (i.e. trends, links to climate forcings and drivers), multi-decadal datasets are required to allow meaningful conclusions.

9. Current research perspectives

9.1. Model development and uncertainties

As we highlighted in this review, while some basic common mechanisms can be identified from multi-model studies, individual models do show a wide range of sensitivity of climate to soil moisture (e.g. Koster et al., 2004a). In particular, there are significant divergences between models regarding soil moisture–precipitation feedbacks (Section 5.2), but also regarding the representation of evapotranspiration (e.g. Guo et al., 2006a; Gedney et al., 2006; Piao et al., 2007; Boé and Terray, 2008) or soil moisture persistence patterns (e.g. Seneviratne et al., 2006b).

One may in particular note a striking issue: Despite a strong increase in the complexity of the available land surface models (Section 4.2), it is apparent that even simple relationships and parameters are still relatively unknown. For instance, the water-holding capacity, which was already recognized as one of the key parameters for soil moisture–climate relationships in early studies (e.g. Delworth and Manabe, 1988; Milly and Dunne, 1994), is highly variable between state-of-art AGCMs (Seneviratne et al., 2006b). Similarly, the dependency of evapotranspiration on soil moisture, which can be considered as the absolute first-order control on most inferred soil moisture–climate feedbacks (Sections 4 and 5) is highly variable among LSMs (Guo et al., 2006a,b; Dirmeyer et al., 2006a,b). Regarding this latter point, it is important to note that the highlighted agreement between models and observations (Fig. 8 and Sections 4 and 5) is mostly qualitative rather than quantitative. Large discrepancies are found in many regions, e.g. in the Sahel, where an analysis of evapotranspiration decay time scales (Teuling et al., 2006, see Appendix B.3.) suggests an underestimation of rooting depth and associated land surface memory in current LSMs.

Similarly, very recent results of the LUCID (Land Use and Climate IDentification of robust impacts) project (Pitman et al., 2009) have shown that climate models often follow different assumptions regarding the physical characteristics associated with a specific land cover or land use type, resulting in divergent outcomes regarding the impact of land use change scenarios. Thus, despite several recent developments on the inclusion of physiological processes in LSMs, it appears that biophysical processes, especially those related to soil moisture–vegetation relationships, require more attention as well as a more adequate validation.

Research advances in this area will require a better use of newly available observations (FLUXNET measurements, satellite data), but also more extended model intercomparison studies similar to the PILPS (e.g. Henderson-Sellers et al., 1995, 1996), GLACE/GLACE-2 (Koster et al., 2004a, 2006; Guo et al., 2006a; Koster et al., 2010), GSWP/GSWP2 (Dirmeyer et al., 1999, 2006a), and LUCID (Pitman et al., 2009) initiatives. This is in particular the case regarding soil moisture–climate interactions under enhanced greenhouse gas concentrations.

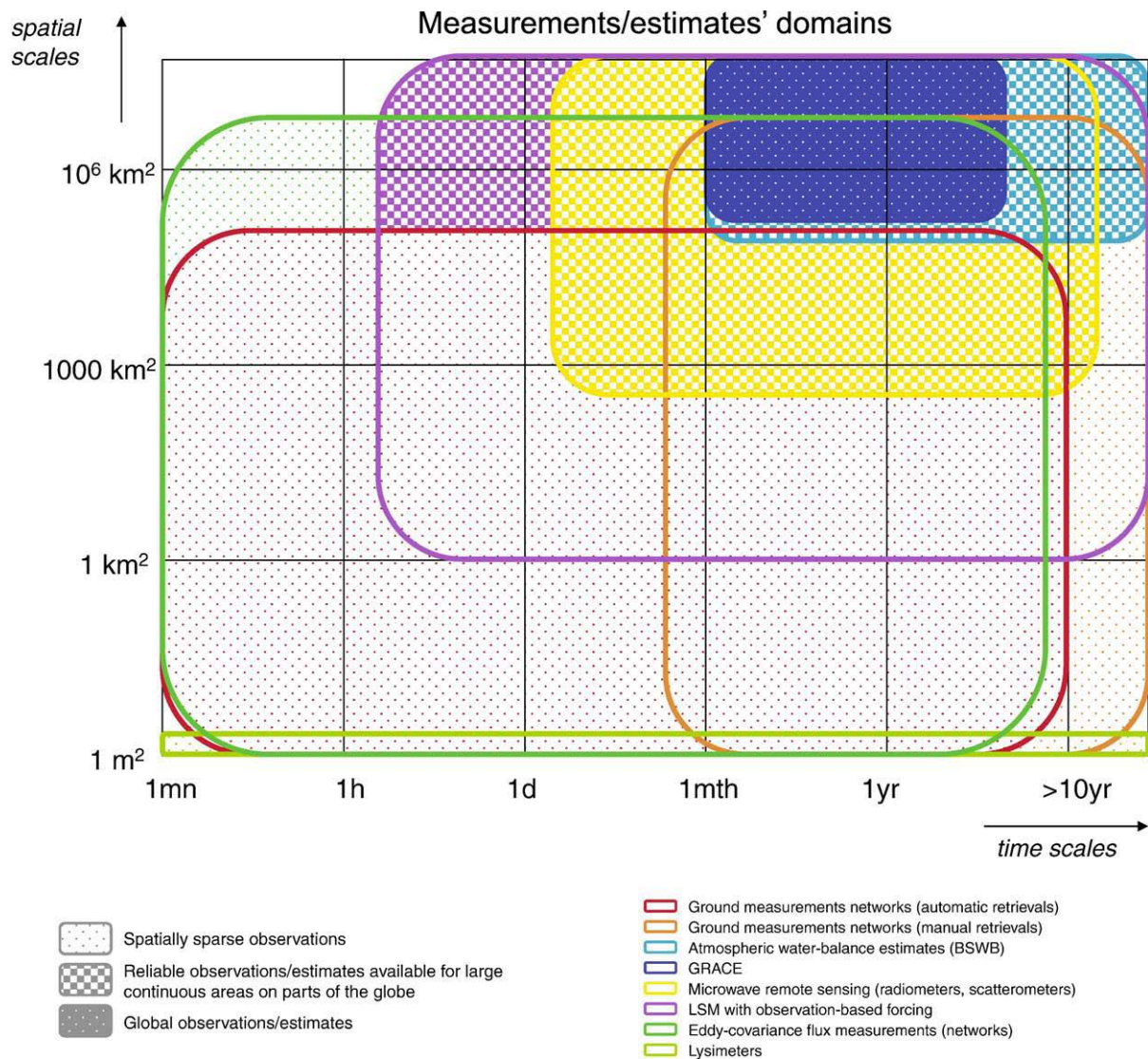


Fig. 16. Spatial and temporal scales of measurements/estimates for soil moisture (ground measurements, microwave remote sensing, and LSMs' estimates), terrestrial water storage (atmospheric water balances/BSWB and GRACE), and evapotranspiration (LSMs' estimates, atmospheric water balances/BSWBs, eddy-covariance flux measurements and lysimeters). See also Tables 1 and 2, and Fig. 15.

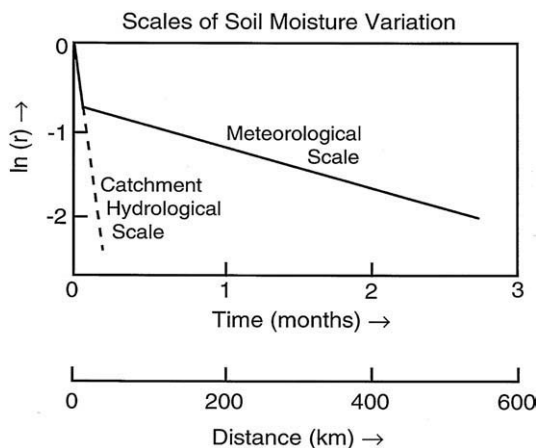


Fig. 17. Schematic diagram of characteristic temporal and spatial scales of soil moisture autocorrelation on the catchment hydrological scale and meteorological scale (from Robock et al., 1998). r is the autocorrelation function. The time and length scales are determined by the slope of the lines. Note that the exact values may differ in the field (see e.g. Entin et al., 2000). [Figure reprinted from Robock et al., 1998, © Elsevier].

9.2. New observational datasets and validation approaches

As mentioned in Section 8, an important limitation for improving our understanding of soil moisture–climate interactions is the lack of comprehensive soil moisture and evapotranspiration observational datasets. While new datasets are becoming available, in particular satellite soil moisture datasets (Section 8.2), the new measurements of the FLUXNET network (Section 8.6), and for instance the water-balance estimates of the BSWB dataset (Section 8.3), these measurements or estimates are often confined to a limited part of the globe, and have distinct temporal and spatial coverage and resolution (Section 8.7). Moreover, even where observational data are available, it is not always straightforward to use them in an effective way to infer land–climate interactions, due to methodological issues (see Section 5.2.4 and Appendix B).

The first challenge for research in this area is to intercompare the available datasets and define new diagnostics (e.g. Appendix B) for the validation of climate and land surface models. The search for new diagnostics is motivated by the fact that the performance of a climate model cannot be reduced to the representation of absolute evapotranspiration, sensible heat flux, precipitation, temperature, or soil

moisture, since for many applications it is more relevant that the coupling between these climate variables is correctly represented. Examples of research in this direction have been provided by the work of Alan Betts (e.g. Betts et al., 1996; Betts, 2004; Betts et al., 2006; Betts, 2007) for single grid points and/or river basins. In Betts (2004), for instance, several tight coupling relationships between the soil moisture index and boundary-layer structure, cloud cover and radiation are highlighted for the ERA-40 reanalysis in selected river basins. A recent study for the European continent has shown that this approach can be well extended to identify and validate the geographical location of soil moisture regimes in a climate model (Jaeger et al., 2009), and Santanello et al. (2009) have also illustrated the use of this methodology for the intercomparison of planetary boundary-layer schemes and land surface models. Similarly, persistence patterns in soil moisture and evapotranspiration (Seneviratne et al., 2006b; Teuling et al., 2006), or the correlation of evapotranspiration with temperature (Seneviratne et al., 2006a), can be used as indirect estimates of soil moisture–evapotranspiration coupling, that can be applied both to observational and model data (see Appendix B for an overview). One particularly challenging issue is the development of diagnostics providing insight on soil moisture–precipitation feedbacks. As mentioned in Section 5.2.4 Appendix B, there are important methodological limitations with the establishment of such relationships, most importantly because of confounding effects induced by the impact of precipitation on soil moisture and the influence of external variables.

A second challenge regarding validation datasets is to attempt to merge the information scattered in single datasets through land data assimilation (see also Section 8.7). Until now, studies generally focused on the assimilation of only one observational soil moisture dataset (sometimes in combination with other land variables, e.g. Balsamo et al., 2007; Muñoz Sabater et al., 2008). However, approaches considering the assimilation of several soil moisture datasets could possibly better make use of their complementarities. Research in the area of land data assimilation is still at its beginning (see also Reichle, 2008), but it could help produce more comprehensive validation datasets for soil moisture and evapotranspiration.

Last but not the least, a remaining key challenge is to continue to develop ground-based monitoring networks for soil moisture and evapotranspiration. Several efforts have been recently initiated in this direction: For example, the soil moisture measurements of the Oklahoma Mesonet (Basara and Crawford, 2000), the REMEDHUS network (Martinez-Fernandez and Ceballos, 2003), the AMMA project (Redelsperger et al., 2006), the SMOSREX ground-measurement experiment (De Rosnay et al., 2006), the NAFE campaign (Panciera et al., 2008), as well as the new TERENO (<http://www.tereno.net/>), SwissSMEX (<http://www.iac.ethz.ch/url/research/SwissSMEX>), and ICOS (<http://www.icos-infrastructure.eu/>) initiatives, to mention just a few. These efforts should be strongly encouraged. Several studies have shown that indirect estimates based on satellite data, diagnostic approaches, and/or modelling approaches need to be evaluated with ground observations (e.g. Seneviratne et al., 2004; Reichle et al., 2004; Hirschi et al., 2006a,b; Guo et al., 2007; Rüdiger et al., 2009), and that their performance may depend on the climate regime and land cover characteristics for instance. Thus, more extended ground observations (together with targeted field experimentation), will be essential for improving our understanding of soil moisture–climate interactions, especially in regions where data coverage is still poor (e.g. South America, Africa, but also Europe in the case of soil moisture).

9.3. Research applications: Subseasonal and seasonal forecasting, numerical weather prediction

Several studies have investigated the potential of soil moisture for subseasonal and seasonal climate predictability (e.g. Fennessy and Shukla, 1999; Koster et al., 2000; Koster et al., 2004a,b; Ferranti and Viterbo, 2006; Kanae et al., 2006; Conil et al., 2007; Koster et al., 2010,

see also Section 6.1). As mentioned, it can be expected that prediction skill will be large in regions with long soil moisture memory and strong soil moisture–climate interactions (e.g. Seneviratne et al., 2006b). Note, however, that land–atmosphere interactions may also have an indirect effect on climate predictability, for instance by enhancing the effects of initial SST anomalies (e.g. Schubert et al., 2004, 2008).

Subseasonal and seasonal forecasting skill associated with soil moisture could be critical for several applications, in particular agriculture. It is thus important to sustain research in this direction, and in particular to investigate the potential of existing soil moisture datasets for soil moisture initialization. The second phase of the Global Land Atmosphere Coupling Experiment (GLACE-2, <http://gmao.gsfc.nasa.gov/research/GLACE-2/>), which is currently on going (Koster et al., 2010), will allow the derivation of multi-model estimates of the subseasonal and seasonal forecasting skill associated with realistic soil moisture initialization.

In principle, soil moisture initialization could also be relevant for numerical weather predictions, given the inferred impacts of soil moisture for temperature and precipitation. These aspects have not been investigated in comprehensive multi-model and long-term experiments so far, though a certain number of single-model studies did consider this issue, and reported forecasting improvements in some (but not all) considered cases either when modifying the soil moisture initial conditions or using a different land surface scheme (e.g. Beljaars et al., 1996; van den Hurk et al., 1997; Betts et al., 1997; Cassardo et al., 2002; De Goncalves et al., 2006; Scipal et al., 2008). Given the projected changes in soil moisture–atmosphere coupling in some regions (e.g. Central Europe, see Section 7), the sensitivity of forecasting skill to soil moisture may change in these areas in the course of the coming decades.

Finally it is important to note that soil moisture by itself is an essential quantity to forecast independently of its impacts on temperature and precipitation, since it is directly related to agricultural drought (Heim, 2002). Early warning systems for drought can make use of the intrinsic memory associated with soil moisture content, beside additional skill that may be gained from the accurate representation of soil moisture–atmosphere feedbacks.

10. Summary and conclusions

We have presented recent developments in the field of soil moisture–climate interactions. While the first studies investigating possible effects of soil moisture on climate date back to the 70s and 80s (e.g. Charney, 1977; Shukla and Mintz, 1982; Yeh et al., 1984), major advancements have occurred in very recent years. First, several multi-model studies have been performed to better assess land–climate interactions independently of single-model effects (e.g. Henderson-Sellers et al., 1995, 1996; Dirmeyer et al., 1999; Koster et al., 2004a; Dirmeyer et al., 2006a,b; Koster et al., 2006; Guo et al., 2006a; Seneviratne et al., 2006a; Pitman et al., 2009; Koster et al., 2010). Second, new validation datasets are becoming available thanks to recently initiated soil moisture and evapotranspiration measurement networks (e.g. Robock et al., 2000; Basara and Crawford, 2000; Baldocchi et al., 2001, Martinez-Fernandez and Ceballos, 2003; De Rosnay et al., 2006; Panciera et al., 2008; Baldocchi, 2008), new satellite measurements (e.g. Tapley et al., 2004; Wagner et al., 2007; Owe et al., 2008; de Jeu et al., 2008), and newly investigated diagnostic approaches (e.g. Seneviratne et al., 2004; Hirschi et al., 2006a,b; Mueller et al., in press).

As a first-order effect, we have seen that one can understand the impact of soil moisture on climate by considering that its effect is strongest in transitional regions between dry and wet climates, and approximately follows the relationship depicted in Fig. 5 (see Section 4). This allows the description of many of the inferred feedbacks in a conceptual framework both in present and future climate (e.g. Sections 4–7). However, while this relationship appears to well represent the qualitative behaviour of both models and observations (Section 4), significant uncertainties remain regarding

quantitative aspects (strength of coupling, depth of storage, and persistence characteristics). These uncertainties should be addressed with observations, to allow a better quantitative representation of this key relationship. Moreover, in order to reach a more thorough understanding of the role of soil moisture within the climate system, it appears evident that further relationships, for instance the exact behaviour of vegetation under enhanced CO₂, the effect of nutrient limitation, and processes impacting infiltration are relevant. Boundary-layer feedbacks and non-local effects, as well as geographical variations in soil and vegetation properties are also associated with large uncertainties. Hence the adequate quantification of these various processes and parameters together with their interactions, in particular based on newly available observational datasets (Section 8) and resulting improvements in their representations in current climate models, will be the key in reducing the uncertainty still remaining regarding soil moisture–climate interactions. This is particularly relevant for climate-change projections.

We can identify some key challenges for research in the area of soil moisture–climate interactions in the coming years (see also Section 9), in particular: 1) to further develop methodological approaches to investigate land–climate interactions and feedbacks, in particular related to new diagnostics of land–atmosphere coupling; 2) to optimize the use of existing (observational) information, be it for model evaluation or data merging (e.g. land data assimilation); 3) to foster interdisciplinary research, i.e. at the interface between water, energy, and carbon-cycle applications, as well as between observational (ground and remote sensing) and modelling research communities; 4) to advance progress in key applications such as seasonal forecasting and numerical weather prediction; 5) to further develop ground-based validation networks for soil moisture and evapotranspiration in new settings (e.g. heterogeneous terrain) and regions (e.g. South America and Africa); 6) to further progress in the understanding of soil moisture–climate interactions in the context of climate change, with the aim of better characterizing (and ultimately reducing) the uncertainty of climate scenarios due to the associated feedback processes.

It is important to note that while we strongly highlighted the need for an extension of reference observational datasets in this field, new observations alone will not allow quantum leaps in our understanding of the relevant processes and feedback mechanisms. There are also major methodological issues, in particular regarding the inference of causality in the case of land–atmosphere feedbacks (see also Section 5.2.4 Appendix B). Because of their complexity, models are ideally suited to investigate interactions and feedback processes, and thus major advances will likely only be yielded with the combination of model experiments and the analysis of existing and new observational data.

In conclusion, the investigation of soil moisture–climate interactions is a very vibrant research field, which lies at the interface between several major scientific disciplines, such as climate science, hydrology, atmospheric science, meteorology, soil physics, and ecophysiology. Despite current limitations due to the lack of ground observations in several regions of the globe, major advances can be expected in the coming years. This will require the joint investigation of soil moisture–climate interactions by various disciplines, but also the combination and comparison of modelling and observational datasets, as well as the conducting of numerical and field experiments. We can expect that increased knowledge on the relevant relationships will allow the better constraint of climate-change projections in several regions of the world, as well as the development of relevant applications for society such as improved drought monitoring, numerical weather prediction, and seasonal forecasting.

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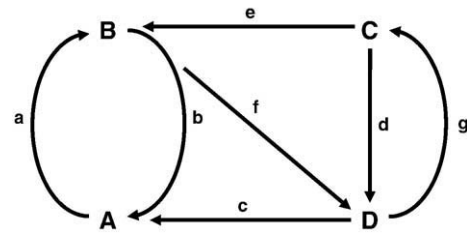


Fig. 18. Definitions of coupling, feedbacks, and interactions (see text). The upper-case letters A, B, C, and D refer to related states or processes, and the lower-case letters a–g refer to the corresponding relationships between the variables A–D.

comments on the manuscript. We also thank Randy Koster for sharing the data of Fig. 8c,d, and Richard de Jeu for sharing the data of Fig. 15c, as well as Esther Jampen for formatting the references of the manuscript.

Appendix A. Coupling, feedbacks, interactions: Terminology

The terms coupling, feedbacks and/or interactions are often used interchangeably in the literature, which can lead to some confusion. Moreover, they can have slightly different definitions depending on the context, e.g. modelling framework or physical description. Hence, we define here more precisely what is meant with each of these terms in this review using the schematic in Fig. 18. This schematic displays the interactions taking place between four climate variables (A, B, C, and D). The (arbitrarily defined) relationships between the variables are referred to with the lower-case letters a) to g).

Whenever we refer to coupling in this article, we mean the degree to which one variable controls another one (for instance relationships a) and b) in Fig. 18. Relationship a) in Fig. 18 refers to the coupling of B with A, i.e. A–B coupling, while relationship b) refers to the coupling of A with B, i.e. B–A coupling. The relationships a) and b) refer to a one-way coupling. In case of land–atmosphere interactions, A could for instance represent soil moisture and B air temperature, whereby a) represents the control of soil moisture on temperature (soil moisture–temperature coupling), and b) represents the control of temperature on soil moisture (temperature–soil moisture coupling). When A is coupled to B and B is coupled to A as well, we can refer to the resulting relationship as a two-way coupling (or feedback, see below). By extension, coupling is not only defined between states and/or processes (e.g. soil moisture and temperature), but also at interfaces between spheres (land and atmosphere). In the latter case, it refers to the extent to which processes in one sphere (e.g. land) can impact states or processes in another sphere (e.g. atmosphere). While coupling between single variables can be easily quantified in a modelling framework (e.g. GLACE experiment; see also Appendix B.1), coupling at interfaces is not a well-defined quantity, and is loosely used to collectively refer to the main coupling relationships between states and processes in the two considered spheres.

The term coupling is based on the analogy with a modelling framework, where e.g. land and atmospheric modules can be used either in coupled or uncoupled modes. The coupled mode refers to two-way coupling between the land and atmospheric modules, while the uncoupled mode refers to simulations where one or the other component is prescribed, i.e. one-way coupling only (e.g. forcing of a land surface model with output from an atmospheric model, or forcing of an atmospheric model with prescribed land surface conditions). The comparison between coupled and uncoupled simulations allows to quantify the impact of the full two-way coupling, and thus of one of the two coupling directions (e.g. soil moisture–temperature coupling).

A two-way coupling, such as in the case of the relationships a) and b) is more commonly referred to as feedback. For instance the term “soil moisture–precipitation feedback” refers to the possible retroaction of soil moisture on itself through the combination of its impact on precipitation, and of the return impact of precipitation on soil moisture.

Finally we use the term “interactions” in a more general sense, for instance to refer to the sum of relationships between states and processes at the land–atmosphere interface (e.g. sum of relationships a–g in Fig. 18) without indication of the direction of causality.

Appendix B. Coupling diagnostics

In this appendix, we present a few approaches commonly used to quantify coupling between processes at the land–atmosphere interface, either based on numerical experiments or on the analysis of model and observational data.

B.1. Coupling diagnostics based on numerical experiments

In a modelling framework, numerical experiments can be used to diagnose the sensitivity of climate processes to land surface conditions. A typical example is the early study of Shukla and Mintz (1982), which investigated the impact of dry and wet land surface conditions for climate, by specifying evapotranspiration equal to zero or potential evaporation, respectively. The net effect of the implemented change is estimated by comparing the sensitivity experiments with the control experiment for certain climate variables. For instance, if the sensitivity experiment investigates the impact of the mean change ΔA in the variable A (e.g. soil moisture and evapotranspiration) for the variable B (e.g. temperature and precipitation), the sensitivity of B to A can be estimated by analysing $\Delta B = B_{\text{SENS}} - B_{\text{CTL}}$, where the subscripts CTL and SENS refer to the control and sensitivity experiments, respectively. From this, the sensitivity of B to A can be approximated as:

$$\frac{dB}{dA} \approx \frac{\Delta B}{\Delta A} \quad (15)$$

Obviously, Eq. (15) is a rough approximation, given the non-linearity of the investigated processes, but the dependency of B on A can be refined by performing a large number of sensitivity experiments with slightly modified perturbations and climate conditions.

In the context of applications related to seasonal forecasting or the analysis of extreme events, impacts on variability are often more relevant than impacts on mean climate. In this case, another type of sensitivity experiments can be performed, where only the (e.g. interannual and intra-annual) variability of A (e.g. soil moisture) is removed (e.g. Koster and Suarez, 2001; Seneviratne et al., 2006a; Jaeger and Seneviratne, in press). The contribution of the variability (variance) of A to the variability (variance) of B on the given time scale can be estimated as follows:

$$\text{fraction of } \sigma^2(B) \text{ explained by } \sigma^2(A) = \frac{\sigma^2(B)_{\text{CTL}} - \sigma^2(B)_{\text{SENS}}}{\sigma^2(B)_{\text{CTL}}} \quad (16)$$

where the “CTL” subscript refers to the control experiment and the “SENS” subscript to the sensitivity experiment with removed variance of variable A .

A further important coupling diagnostic is the estimation of the potential predictability associated with variable A (again, in principle soil moisture) for a given atmospheric variable B (e.g. temperature, precipitation). The approach followed in the Global Land–Atmosphere Coupling Experiment (GLACE; Koster et al., 2004a, 2006) to determine this quantity is to conduct two ensemble experiments, one with fully coupled variable A /soil moisture (ensemble W), and the other one with prescribed variable A /soil moisture (ensemble S), where the prescribed variable A /soil moisture fields are taken from one simulation of the ensemble W . The potential predictability of the variable B (e.g. precipitation and temperature) in one or the other

ensemble can be estimated using the Ω diagnostic proposed by Koster et al. (2000):

$$\Omega_B = \frac{N\sigma_B^2 - \sigma_B^2}{(N-1)\sigma_B^2} \quad (17a)$$

with

$$\hat{B}(t) = \frac{1}{N} \sum_{i=1}^N B_i(t) \quad (17b)$$

where N refers to the number of ensemble members. The time series $B(t)$ is defined as a series of multi-day averages (e.g. six days).

The Ω diagnostic can be seen as a measure of the coherence of the simulations within a given ensemble (Yamada et al., 2007). By comparing the Ω values for the W (coupled) and S (uncoupled/prescribed) ensembles, an estimate of the maximum potential increase in predictability attainable by prescribing variable A /soil moisture can be defined as follows:

$$\Delta\Omega = \Omega_B(S) - \Omega_B(W) \quad (18)$$

The quantity $\Delta\Omega$ can be thus considered as an indirect measure of coupling focused on (potential) predictability (e.g. Koster et al., 2004a). Note that this framework has until now only been applied to estimate the potential predictability associated with soil moisture, but could be similarly applied for other land variables such as snow.

Coupling diagnostics based on numerical experiments as described in this subsection, have the advantage to be unambiguous in terms of causality, are ideal tools for inter-model comparisons, and can provide global estimates of the given quantities. However, there is one important caveat: Such sensitivity experiments cannot be paralleled in the field and, thus, the derived diagnostics cannot be validated. Moreover, the sensitivity experiments have a high computational cost and require coordination between modelling groups. The GLACE experiment, for instance, did consider a large number of existing AGCMs. Nonetheless, these correspond only to a small fraction of the AOGCMs used for IPCC AR4 projections. Hence, it is useful to identify coupling diagnostics that can be applied both to model and observational data, and do not require the computation of additional numerical experiments.

B.2. Correlation measures and empirical relationships

While correlations cannot demonstrate links of causality, they can nonetheless be used as indirect estimates of coupling for relationships for which causality has been established. As an example, the correlation between temperature and evapotranspiration $\rho_{(E,T)}$ has been shown to be an adequate diagnostic for soil moisture–temperature coupling (Seneviratne et al., 2006a). This is the case both on the seasonal time scale (e.g. JJA) as well as on the monthly time scale (e.g. Jaeger et al., 2009). As illustrated in Fig. 19, regions with soil moisture–limited evapotranspiration are characterized by a negative $\rho_{(E,T)}$, while regions with energy limitation are generally characterized by a positive $\rho_{(E,T)}$ (see also Section 5.1). Regions identified by these means agree well with those derived with other diagnostics regarding soil moisture–temperature coupling (such as Eqs. (16) or (18), see e.g. Section 5.1.2). Another example of the use of correlations as diagnostics is provided in Figs. 8a,b, which display the correlation of evapotranspiration with global radiation ($\rho_{(Rg,E)}$), on yearly, respectively daily, time scales, and in the case of Fig. 8a also with precipitation ($\rho_{(P,E)}$, on yearly time scale), so as to determine regions with moisture-limited vs energy-limited evapotranspiration regimes.

Lag correlation (or covariance) analyses have also been applied to the investigation of soil moisture–precipitation coupling (e.g. Findell and Eltahir, 1997; Notaro, 2008). However, the value of such an

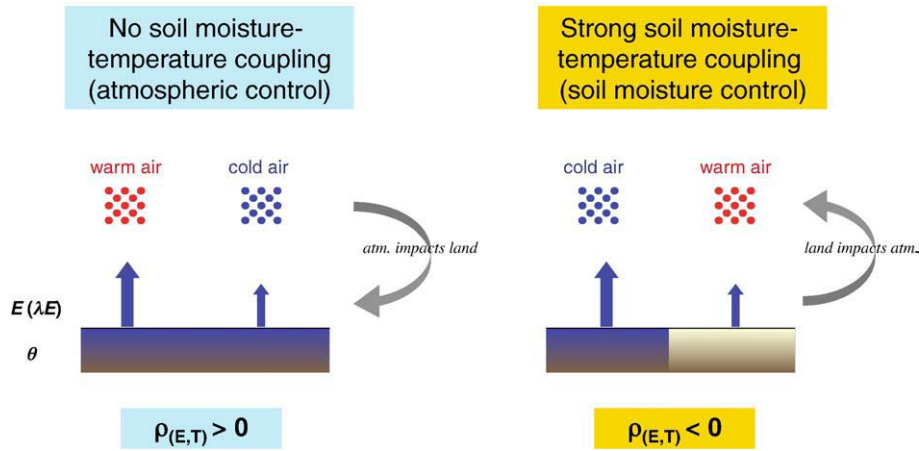


Fig. 19. Illustration of interpretation of $\rho_{(E,T)}$ as diagnostic of soil moisture–temperature coupling.

analysis is strongly limited by the known impact of precipitation on soil moisture. Indeed, lag correlations between soil moisture and precipitation could be linked to autocorrelation features in the precipitation only, without any contribution of soil moisture feed-backs (e.g. Salvucci et al., 2002; Wei et al., 2008; Orlowsky and Seneviratne, *in press*). Moreover, some diagnosed lag correlations can be artifacts of the data handling (e.g. Findell and Eltahir, 1997; Salvucci et al., 2002).

A further common approach to identify coupling between related variables is to derive empirical relationships by displaying the investigated variables as a function of one another (e.g. Betts, 2004; Betts et al., 2006; Betts, 2007; Jaeger et al., 2009; Santanello et al., 2009). While again, no exact direction of causality can be established by these means, these relationships can be suggestive of coupling mechanisms at the land–atmosphere interface. They are also of strong relevance for model validation, since they can be derived both for observations and model data.

B.3. Derivation of evapotranspiration sensitivity to soil moisture from its decay time

Negative feedbacks lead to an exponential decay of initial anomalies. This is the case for the soil moisture–evapotranspiration relationship under dry-downs (Teuling et al., 2006). Under the assumption that evapotranspiration linearly depends on soil moisture in this regime, i.e.;

$$E(t) = cS(t) \quad (19)$$

where $E(t)$ is evapotranspiration as function of time and $S(t)$ is absolute soil moisture as function of time, and when soil moisture is only modified by evapotranspiration (i.e. rain-free periods with no drainage), it can be shown that the decay time scale is equal to $1/c$. This allows the empirical derivation of the sensitivity factor c from evapotranspiration measurements (see Teuling et al., 2006). While models and observations display consistent decay time scales in some regions, large discrepancies were identified in the Sahel region, possibly due to the underestimation of root depth (e.g. Schenk and Jackson, 2005).

Acronyms and symbols

λ	Latent heat of vaporization
$\rho_{(E,T)}$	Correlation between evapotranspiration and temperature (coupling diagnostic) – see Appendix B.2.
$\rho_{(R_g,E)}$	Correlation between evapotranspiration and global radiation – see Section 4.4 and Appendix B.2.
$\rho_{(P,E)}$	Correlation between evapotranspiration and precipitation – see Section 4.4 and Appendix B.2.

Ω	Omega diagnostic (measure of coherence within ensemble of simulations) – see Appendix B.1.
$\Delta\Omega$	Delta omega diagnostic (coupling diagnostic derived from difference in coherence in ensembles of coupled and uncoupled simulation) – see Appendix B.1.
θ	Soil moisture (volumetric).
θ_{CRIT}	Critical soil moisture content (under which soil moisture becomes limiting for evapotranspiration) – see Sections 4.1 and 4.2 and Eq. (8b)
θ_{FC}	Soil moisture content at field capacity – see Section 2
θ_{SAT}	Saturation soil moisture content (or water-holding capacity) – see Section 2
θ_{WILT}	Wilting point (also called permanent wilting point or plant wilting point) – see Section 2
20C3M	Simulations from IPCC AR4 for 20 th century (control)
AGCM	Atmospheric General Circulation Model
AMSR-E	Advanced Microwave Scanning Radiometer–Earth Observing System
AOGCM	Atmosphere–Ocean General Circulation Model
ASCAT	Advanced SCATterometer
BSWB	Basin-Scale Water-Balance dataset – see Section 8.3
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA40	ECMWF ReAnalysis–40
ERS	European Remote Sensing satellite
ESA	European Space Agency
EU-FP6	Sixth Framework Program of the European Union
EUMETSAT	European organisation for the exploitation of METeoro-logical SATellites
GCM	General Circulation Model
GLACE	Global Land–Atmosphere Coupling Experiment
GLACE-2	2nd Phase of the Global Land–Atmosphere Climate Experiment
GLDAS	Global Land Data Assimilation System
GRACE	Gravity Recovery and Climate Experiment – see Section 8.2.2
GPCC	Global Precipitation Climatology Centre
GPCP	Global Precipitation Climatology Project
GSWP	Global Soil Wetness Project
GSWP2	2nd Phase of the Global Soil Wetness Project
IPCC	Intergovernmental Panel on Climate Change
IPCC AR4	4th Assessment Report of the Intergovernmental Panel on Climate Change
IPCC AR5	5th Assessment Report of the Intergovernmental Panel on Climate Change
JJA	June–July–August (boreal summer)
LAI	Leaf Area Index
LSM	Land Surface Model
LUCID	Land Use and Climate: IDentification of robust impacts

MetOp	Meteorological Operational satellite programme (jointly established by ESA and EUMETSAT)
NASA	National Aeronautics and Space Administration
PAR	Photosynthetically Active Radiation
PDSI	Palmer Drought Severity Index
PILPS	Project for the Intercomparison of Land Parameterization Schemes
PRUDENCE	Prediction of Regional scenarios and Uncertainties for Defining European Climate change risks and Effects (EU-FP6 project)
RCM	Regional Climate Model
SMAP	Soil Moisture Active and Passive mission
SMI	Soil Moisture Index
SMOS	Soil Moisture and Ocean Salinity mission
SMMR	Scanning Multichannel Microwave Radiometer
SPI	Standardized Precipitation Index
SRES	IPCC Special Report on Emissions Scenarios
SRES A1B	SRES scenario A1B
SRES A2	SRES scenario A2
SST	Sea Surface Temperature
TDR	Time Domain Reflectometry
TMI	TRMM Microwave Imager
TRMM	Tropical Rainfall Measuring Mission

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